

Vertical critical ideals and cyclotomic first jets of q-Painlevé I

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Abstract

We determine the complete first-layer critical ideal and the higher cyclotomic first jets of the divided state differential of the original q-Painlevé I invariant. Reduction of this differential is a Hasse power times the small-order invariant differential, but that residue does not determine its tangent component on a supersingular level. We identify this component through the actual Bockstein for the relevant anticanonical power, an integer prime-trace congruence, and the connecting class of locally exact differentials on the complete fiber. For every prime and every tame block size, near every complete smooth finite fiber, the full first-layer coefficient ideal is generated by the cyclotomic parameter and the Hasse polynomial. At higher odd-prime levels we obtain a complete first-jet factorization and its exact truncated ideal, retaining multiple Hasse roots and arbitrary unit-time jets. In characteristic two the comparison starts at height two: a mixed cubic relation gives a truncated transverse quotient of length five with a specified action of the base parameter. We prove the actual spectral Jacobian and closed-fiber Hasse identifications needed to interpret these statements on the original eight-center surface, including its four terminal affine lines. Higher full thickness and singular-level critical ideals are not determined.

1 Introduction and main results

An invariant can become a pure power after reduction even though its state differential, divided integrally before reduction, retains nonzero information. For the root-of-unity q-Painlevé I invariant, that divided differential reduces to a Hasse power times the small-order invariant differential. This residue describes ordinary levels but vanishes on supersingular levels. We determine the missing vertical information: the complete coefficient ideal at the first cyclotomic layer and the full first jet at every higher layer.

The first-layer result is controlled by a specific obstruction. The original small-order invariant is a section of an anticanonical power, and this section does not lift across the first infinitesimal cyclotomic thickening. An integer congruence for the original matrix trace turns its Bockstein into the connecting class of a locally exact differential on each complete supersingular fiber. This differential is nowhere zero. It supplies the second state direction that is absent from the residue formula and determines the full local ideal. Higher layers retain a common factor at odd primes, whereas characteristic two has a different height-two base and a mixed cubic relation.

1.1 The original invariant and its model

We use the q-Painlevé I coordinates and root-of-unity integral of Joshi–Roffelsen [11]. The one-step rational map is

$$F_t(x, y) = \left(\frac{t}{x - y/s}, \frac{sx}{y} \right), \quad t' = st.$$

All parameters called time below are units. The invariant is fixed by the original matrix, in the auxiliary normalization $w = 1$. Put $b = x(y - 1)$, $d_0 = y - 1$, and $J_1(t) = y - x + x/y - t/x$. Then

$$A(z; t) = \begin{pmatrix} t - b & -x \\ d_0(b - t) & b \end{pmatrix} + z \begin{pmatrix} J_1(t) - 1 & 1 \\ J_1(t) + b - 1 & 1 \end{pmatrix} + z^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (1.1)$$

Its determinant is z^3 . If s has exact order r , the descending product and the integral are

$$M_{r,s}(z) = A(s^{r-1}z; t) \cdots A(z; t), \quad I_{r,s} = [z^r] \operatorname{tr} M_{r,s}(z). \quad (1.2)$$

The original identities are

$$\operatorname{tr} M_{r,s} = t^r + I_{r,s} z^r + z^{2r}, \quad \det M_{r,s} = (-1)^{r+1} z^{3r}, \quad I_{r,s} = -t^r x^{-r} + O(x^{-r+1}). \quad (1.3)$$

Here the last expression is a Laurent valuation statement, not an analytic asymptotic. These matrix and integral identities are source inputs, not new results of this paper; the formula numbering of the author version specified in the bibliography is used when needed.

Let S be the original eight-center blowup of $\mathbb{P}^1 \times \mathbb{P}^1$, let $D = -K_S$ be its boundary cycle, and put $\mathcal{U} = S \setminus D$ and $L_n = \mathcal{O}_S(nD)$. Section 2 gives the centers over the integer parameter ring and their terminal charts. The open surface contains the torus and four complete terminal affine lines. None of the results below removes these lines. We retain the original time, energy, and two relative state directions; replacing the matrix by an isospectral one would not define the same problem.

1.2 Cyclotomic parameters and the three results

Fix a prime p , an integer $m \geq 1$ prime to p , and $a \geq 1$. Let $\mathcal{O}_0$ be an unramified p -adic discrete valuation ring containing a primitive m th root $\tilde{\eta}$, with perfect residue field k . Set

$$\mathcal{O}_a = \mathcal{O}_0[\zeta_{p^a}], \quad \pi_a = \zeta_{p^a} - 1, \quad s_a = \tilde{\eta} \zeta_{p^a}, \quad t_a \in \mathcal{O}_a^\times, \quad N = p^a.$$

On the same eight-center model with $q = s_a$ and time t_a , define

$$\alpha_{mN} = p^{-a} d_{\text{state}} I_{mN, s_a}. \quad (1.4)$$

The differential fixes the time and the base ring. Its division occurs in characteristic zero; Proposition 5.3 proves its regularity on the entire $\mathcal{U}_a$. With $\eta = \bar{s}_a$, put

$$J = I_{m,\eta}(x, y; \bar{t}_a), \quad T = \bar{t}_a^m, \quad \varepsilon_m = (-1)^{m+1}, \quad \sigma = \frac{p^a - 1}{p - 1}.$$

The polynomial relevant to the complete original fibers is

$$H_p(T, h; \varepsilon) = \begin{cases} [Z^{p-1}]((T + hZ + Z^2)^2 - 4\varepsilon Z^3)^{(p-1)/2}, & p \neq 2, \\ h, & p = 2. \end{cases} \quad (1.5)$$

Write $H = H_p(T, J; \varepsilon_m)$. It is monic of degree $p - 1$ in the energy variable, but its roots are not assumed simple. Sections 3–4 prove that a complete smooth finite fiber $X = (J = h)$ of the original pencil is supersingular exactly when $H_p(T, h; \varepsilon_m) = 0$.

For a regular one-form γ on a smooth relative surface, $\mathfrak{c}(\gamma)$ denotes the ideal generated by both coefficients in a local frame of the relative cotangent sheaf. An invertible change of frame does not change this ideal. The notation $\gamma^{[2]}$ means reduction modulo π_a^2 , whose parameter image we denote by ϵ . If that quotient has characteristic p and $b > 0$ with $p \mid b$, the b th power of any lift of a residue function is independent of the lift: changing it by ϵu changes the power by $b\epsilon u$ times the preceding power, which is zero. We write this well-defined lift-power as $f^{(b)}$; the zero power is one.

Theorem 1.1 (All first layers). *Let $a = 1$, with k perfect. At every point of every original complete smooth finite fiber, including its terminal points,*

$$\mathfrak{c}(\alpha_{mp}) = (\pi_1, \tilde{H}), \quad (1.6)$$

for any local lift $\tilde{H}$ of H . If h is a Hasse root of multiplicity e_h , the completed ideal is (π_1, z^{e_h}) , where $\bar{z} = J - h$, after an allowed unramified residue-field extension.

On each complete smooth finite fiber, the actual Bockstein and the restriction trivialized by the original section 1 satisfy

$$0 \neq \kappa_J = \rho_X \beta_{L_m}(J) \in H^1(X, \mathcal{O}_X).$$

On a smooth supersingular fiber, the first tangent differential ν constructed from the original trace satisfies the exact comparison

$$\partial\nu = \text{Fr}_* \kappa_J \neq 0 \quad \text{in } H^1(X, \mathcal{O}_X^p).$$

The Frobenius here maps to its image sheaf, not to $H^1(X, \mathcal{O}_X)$ after the subsequent inclusion.

The complete proof is in Section 6. In particular, (1.6) is an equality of the original full ideals, not just of their images modulo π_1^2 .

Theorem 1.2 (Odd-prime higher first jets). *Let k be finite, p odd, and $a \geq 2$. Identify $\mathcal{O}_1/(\pi_1^2)$ and $\mathcal{O}_a/(\pi_a^2)$ with $k[\epsilon]/(\epsilon^2)$ by $\pi_1 \mapsto \pi_a$, and choose t_1 with the same full image as t_a in this ring. On the entire common open model,*

$$\alpha_{mp^a}^{[2]} = H^{\langle \sigma-1 \rangle} \alpha_{mp}^{[2]}. \quad (1.7)$$

Near every original complete smooth finite fiber,

$$\mathfrak{c}(\alpha_{mp^a}) + (\pi_a^2) = (\pi_a^2, \tilde{H}^\sigma, \pi_a \tilde{H}^{\sigma-1}). \quad (1.8)$$

At a Hasse root of multiplicity e_h , its completed form is $(\pi_a^2, z^{e_h \sigma}, \pi_a z^{e_h(\sigma-1)})$.

The ring identification in this statement is not the natural embedding of cyclotomic fields. Matching only the residue of the time is insufficient; the complete time jet is matched. The form identity does not require a smooth level or a nonzero Hasse value. The ideal assertion uses smoothness and Theorem 1.1 after that identity has been proved in Section 7.

Theorem 1.3 (Characteristic-two higher first jets). *Let $p = 2$, m be odd, and $a \geq 2$, with $N = 2^a$. Identify $\mathcal{O}_2/(\pi_2^2)$ and $\mathcal{O}_a/(\pi_a^2)$ with $B = k[\epsilon]/(\epsilon^2)$ by $\pi_2 \mapsto \pi_a$, and match the complete images of t_2, t_a . Over a perfect residue field, on the entire common open model,*

$$\alpha_{mN}^{[2]} = J^{\langle N-4 \rangle} \alpha_{4m}^{[2]}. \quad (1.9)$$

For finite k , near every original complete smooth finite fiber,

$$\mathfrak{c}(\alpha_{mN}) + (\pi_a^2) = (\pi_a^2, j^{N-1} + \pi_a \tilde{T} j^{N-4}, \pi_a j^{N-2}), \quad (1.10)$$

where j is any local lift of J and $\tilde{T}$ a constant unit lift of T . At $a = 2$ on $X = (J = 0)$, the completed truncated quotient, after the allowed unramified extension, is

$$\frac{k(P)[[w, z, \epsilon]]}{(\epsilon^2, z^3 + \epsilon T, \epsilon z^2)} \simeq \frac{k(P)[[w, z]]}{(z^5)}, \quad \pi_2 \mapsto z^3/T, \quad (1.11)$$

where w is a parameter along the curve and $\bar{z} = J$.

The characteristic-two comparison starts at height two. The height-one ring $\mathcal{O}_1/(\pi_1^2)$ has characteristic four and cannot be used as its dual-number base. Section 8 computes the actual four-block trace and its tangent class before eliminating the two local coefficients. The integer five in (1.11) is a transverse length of the truncated quotient: the parameter w remains, and no length of the original full critical quotient is asserted. After a finite unramified extension a cube root of T normalizes its unit coefficient; T is not a new formal modulus.

A state lift means a point over the same cyclotomic ring after a finite unramified extension of its unramified coefficient ring. The common order of a form is the minimum valuation of its two pulled-back state coefficients, not the differential of the constant lifted point. At ordinary smooth states, α_{mp^a} has order zero and dI_{mp^a} has order $a\varphi(p^a)$. At supersingular smooth states the exact values and lower bounds are as follows.

Cyclotomic layer	Order of α_{mp^a}	Order of dI_{mp^a}
Any p , $a = 1$	1	p
Odd p , $a \geq 2$	≥ 2	$\geq a\varphi(p^a) + 2$
$p = 2$, $a = 2$	1	5
$p = 2$, $a \geq 3$	≥ 2	$\geq a2^{a-1} + 2$

Table 1: Orders along the allowed unramified smooth supersingular states. Inequalities are not claimed to be equalities.

1.3 Relation to existing mechanisms and scope

The initial-value surface and its eight-point geometry originate in the q-Painlevé literature [10, 11]. Anticanonical-cycle periods and the relation between a normal-bundle torsion order and a Halphen pencil are established tools [4, 8]. We need their values on the original integer charts, and must separately exclude quasi-elliptic behavior in the allowed small characteristics. Spectral line-bundle correspondences and cyclic Picard constructions are likewise classical [2, 9]. The proof here supplies the actual state inverse, excludes a hidden inseparable degree or pullback kernel, and descends the torsor action over the original field. Minimal-model uniqueness [15, Tags 0C6B and 0C9Z] is applied only after both models over the same local energy ring have been shown minimal.

Hasse powers and semilinear iteration are not new mechanisms. Cartier conventions require care even for one-dimensional spaces [1]. Vlasenko’s higher Hasse–Witt congruences already include the residue iteration, and the formal-group integrality statement in the cited author version is not restricted to ordinary fibers [16, Theorems 1(i) and 2, author version v3]. Our comparison uses that specified version; no theorem from it is a proof premise below. Dwork-crystal and higher-Hasse constructions give finer congruences under their respective invertibility and Frobenius-lift hypotheses [3, 17]. We prove the needed coefficients and all block deformations directly, including the supersingular locus and characteristic two.

The local division of Frobenius differences and the resulting obstruction classes have a broad prior theory [5–7]. Recent Frobenius–Witt formulations also treat arithmetic Kodaira–Spencer classes [14]. These theories do not by themselves identify the specific L_m section with the original trace congruence used here. Nor is ramification alone a distinction from that literature. Root-of-unity first nonzero ramified terms occur in other specified systems, for example in quantum K-theoretic spectral relations [12]; a spectral relation is not an identity of the present two-direction critical ideals. Lubin–Tate already give a characteristic-two height-two elliptic lifting family [13, Section 3.5]. Accordingly, neither height two, the number five, nor a normalizable unit parameter

is counted as a separate general mechanism here. The calculation concerns the mixed ideal and the original base-parameter action in this model.

All topic-specific proofs needed for the three theorems are given in the body. Sections 2–4 establish the original geometric and Hasse identifications; Sections 5–6 prove the integral trace and complete first-layer ideal; Sections 7 and 8 give the two higher branches. The paper does not determine higher full thickness, singular-level critical ideals, or exact orders after extra ramified state extensions. It makes no general crystalline or formal-group comparison and no claim about Riemann zeros.

2 The original surface and primitive pencil

The geometry needed below belongs to the original space of initial values, not to a surface chosen from its spectral equation. We first construct it over $R = \mathbb{Z}[q^{\pm 1}, \tau^{\pm 1}]$. Write S/R for the following successive blowups of $\mathbb{P}_R^1 \times_R \mathbb{P}_R^1$; x, y denote the coordinates on its original torus. In each row the centers are taken in the indicated order. The last column gives a neighborhood of the terminal exceptional line in the complement of the polar boundary.

	centers	terminal chart	
1	$(x^{-1}, y - 1) = (0, 0)$	$x = u^{-1}, y = 1 + uv$	(2.1)
2	$(x, y^{-1}) = (0, 0), xy = \tau$	$x = u(\tau + uv), y = u^{-1}$	
3	$(x, y) = (0, 0), E_{3,1} \cap \{y = 0\}, x^2/y = \tau$	$x = u(\tau + uv), y = u^2(\tau + uv)$	
4	$(x^{-1}, y^{-1}) = (0, 0), y/x = q$	$x = [u(q + uv)]^{-1}, y = u^{-1}$	

Here a center specified by one displayed ratio is the point on the preceding exceptional divisor where that ratio has the displayed value. The distribution is $1 + 2 + 3 + 2 = 8$. These are the initial-value charts of [11]; the computations below use their actual coordinates. In particular, the two occurrences of τ lie on different components and do not merge a pair of centers.

Let $E_{i,j}$ denote the final strict transform of the j th exceptional curve in cluster i . Order the boundary components as

$$(D_1, \dots, D_8) = (C_{x0}, E_{3,1}, E_{3,2}, C_{y0}, C_{x\infty}, E_{4,1}, C_{y\infty}, E_{2,1}),$$

$$D = \sum_{i=1}^8 D_i. \quad (2.2)$$

Subscripts on this cycle are read modulo eight. Put $\mathcal{U} = S \setminus D$, $L_j = \mathcal{O}_S(jD)$, and $\mathcal{N} = \mathcal{O}_D(D)$. Specialization $q = s, \tau = t$ will be indicated by subscripts only when needed.

Proposition 2.1. *The surface S/R is smooth and projective. Its divisor D is a relative simple-normal-crossing cycle of eight rational curves, $D = -K_{S/R}$, and, on every geometric fiber, $D_i^2 = -2$, $D \cdot D_i = 0$, and $D^2 = 0$. The divisor D is nef. The open surface $\mathcal{U}$ consists of the original torus and the four complete terminal affine lines in (2.1). The form $\omega_S = dx \wedge dy/(xy)$ extends to a nowhere-degenerate relative two-form on $\mathcal{U}$.*

Proof. Each blowup center is a section with two relative coordinates a, b ; its blowup charts have coordinates $(a, b/a)$ and $(a/b, b)$. They are relative affine-plane charts, proving smoothness at each stage, while blowups preserve projectivity. The center in cluster 2 is obtained by writing $y^{-1} = u$, $x = u\xi$ and then $\xi = \tau + uv$. For cluster 3, write $x = a, y = ab$, then $b = u, a = u\xi$, and finally $\xi = \tau + uv$. For cluster 4 write $y^{-1} = u, x^{-1} = u\xi$, then $\xi = q + uv$. Cluster 1 uses $x^{-1} = u, y - 1 = uv$. This verifies every center and chart in (2.1).

One may first perform the four node blowups and then the four non-node blowups. The latter occur at the unit coordinates $1, \tau, \tau, q$, so remain away from the nodes after any base change. At a boundary node a two-form with a simple pole is locally $a_0 da \wedge db / (ab)$, with a_0 a unit. Substitution $a = u, b = uv$ gives $a_0 du \wedge dv / (uv)$: the new exceptional curve remains in the polar boundary. At a smooth boundary point the same substitution sends $a_0 da \wedge db / a$ to $a_0 du \wedge dv$, so the terminal exceptional curve is not polar. No zero divisor is introduced. Consequently $\text{div}(\omega_S) = -D$, giving $D = -K_{S/R}$.

The node blowups insert the four intermediate components into the coordinate rectangle, in exactly the cyclic order (2.2). The vectors of orders of x, y along the intermediate boundary are

$$(1, 0), (1, 1), (1, 2), (0, 1), (-1, 0), (-1, -1), (0, -1), (1, -1).$$

Before the last four blowups its self-intersections in this order are $-2, -2, -1, -2, -1, -1, -2, -1$. A node blowup lowers the self-intersection of each incident curve by one and creates a (-1) -curve; these rules give the displayed list directly from the original rectangle of self-intersection-zero curves. The four final blowups lower each remaining -1 to -2 . Adjacent components meet transversely once and other pairs do not meet. Thus $D \cdot D_i = -2 + 1 + 1 = 0$. An irreducible curve not in D has nonnegative intersection with each effective component of D , which also proves nefness.

Each terminal exceptional curve meets D once, and its complement is the whole line $u = 0$, $v \in \mathbb{A}^1$. Substitution in ω_S , in the order of (2.1), gives

$$-\frac{du \wedge dv}{1 + uv}, \quad \frac{du \wedge dv}{\tau + uv}, \quad -\frac{du \wedge dv}{\tau + uv}, \quad -\frac{du \wedge dv}{q + uv}.$$

Their denominators restrict to $1, \tau, \tau, q$ on the terminal lines. These are units; the remaining points of $\mathcal{U}$ are on the torus. This proves regularity and nondegeneracy everywhere. $\square$

2.1 The exact polar divisor

For the geometric arguments let k be a field, $s, t \in k^\times$, and suppose s has exact order r . In positive characteristic this implies that r is prime to the characteristic. Keep the matrix and integral of (1.1) and (1.2). The original integral construction supplies the Laurent identity, invariance, and leading coefficient

$$\begin{aligned} I_{r,s} &\in k[x^{\pm 1}, y^{\pm 1}], & I_{r,s}(x, y; t) &= I_{r,s}(F_t(x, y); st), \\ F_t(x, y) &= \left(\frac{t}{x - y/s}, \frac{sx}{y} \right), & I_{r,s} &= -t^r x^{-r} + O(x^{-r+1}). \end{aligned} \tag{2.3}$$

The last expansion is an x -adic Laurent valuation over $k(y)$, not an analytic asymptotic statement. These inputs are the original matrix-integral identities of [11], Theorem 3.1 and its proof (also in the author's version 2). They hold over the cyclotomic integer ring after clearing the displayed substitution denominators, so specialize to every allowed characteristic. In particular the leading coefficient is always nonzero.

Lemma 2.2. *On the specified compact surface, $(I_{r,s})_\infty = rD$. The sections $1, I_{r,s}$ of L_r are independent and have no common zero. They define a morphism $f : S_{t,s} \rightarrow \mathbb{P}_c^1$ with $c = I_{r,s}$ and $f^{-1}(\infty) = rD$ as a scheme fiber.*

Proof. All divisor calculations may be checked over an algebraic closure. For completeness, the rational map in (2.3) preserves the two-form used above. With $a = x - y/s$ and $(X, Y) = F_t(x, y)$,

$$\det \frac{\partial(X, Y)}{\partial(x, y)} = \frac{st}{y^2 a}, \quad XY = \frac{stx}{ay},$$

$$(x, y) = \left(\frac{tY}{X(Y-1)}, \frac{st}{X(Y-1)} \right).$$

Thus F_t is birational and $F_t^* \omega_{S_{st,s}} = \omega_{S_{t,s}}$. More particularly, it transports the normalized boundary valuations without ramification. The following calculation records all eight transports. The pair in the last column is a target normal parameter and a target coordinate on its divisor; its leading terms are in the source parameter u , with residue coordinate v .

source (x, y)	target	target coordinates
$C_{x0} : (u, v)$	C_{y0}	$(Y, X) = ((s/v)u, -st/v + O(u))$
$C_{y0} : (v, u)$	$C_{y\infty}$	$(1/Y, X) = (u/(sv), t/v + O(u))$
$C_{y\infty} : (v, u^{-1})$	$E_{3,1}$	$(X, Y/X) = (-stu + O(u^2), -v/t + O(u))$
$E_{3,1} : (u, uv)$	$C_{x\infty}$	$(1/X, Y) = ((s-v)u/(st), s/v)$
$C_{x\infty} : (u^{-1}, v)$	$E_{2,1}$	$(1/Y, XY) = ((v/s)u, st/v + O(u))$
$E_{2,1} : (uv, u^{-1})$	$E_{3,2}$	$(Y/X, X^2/Y) = (-(v/t)u + O(u^3), st^2/v + O(u^2))$
$E_{3,2} : (uv, u^2v)$	$E_{4,1}$	$(1/Y, Y/X) = (u/s, sv/t + O(u))$
$E_{4,1} : ((uv)^{-1}, u^{-1})$	C_{x0}	$(X, Y) = (stv u/(s-v), s/v).$

Every leading normal coefficient is a unit in $k(v)$, and every residue map is nonconstant. For each row, therefore, $v_{D_{i,t}}(F_t^*g) = v_{D_{j,st}}(g)$ for every nonzero target rational function g . The rows form an eight-cycle. Invariance in (2.3) transports the pole order r at C_{x0} to every boundary component. The time becomes $s^j t$ during this transport, but its leading coefficient $-(s^j t)^r$ remains a unit.

It remains to exclude poles on the four terminal curves. This requires actual regular local maps, not only limits on those curves. For chart 1 set $w = 1 + uv$, $a = 1 - uw/s$; the coordinates in target chart 2, whose time is st , are

$$u' = uw/s, \quad v' = \frac{st(w^2 - sv)}{aw^2}.$$

For chart 2 set $b = t + uv$, $a = 1 - su^2b$; the target chart 3 is

$$u' = -uba/t, \quad v' = \frac{st^2(v - 2sub^2 + s^2u^3b^3)}{b^2a^3}.$$

Indeed $ba^2 - t = u(v - 2sub^2 + s^2u^3b^3)$ cancels the apparent denominator u in that expression. From chart 3 to target chart 4,

$$u' = u/s, \quad v' = s\{(s-u)v - t\}/t.$$

Finally, on chart 4, writing $b = s + uv$ gives the target torus coordinates $(X, Y) = (-stb/v, s/b)$ near its generic point. All displayed denominators are units at the respective generic points. On the terminal lines the first three maps have residue coordinates $st(1 - sv)$, sv , and $s(sv - t)/t$, respectively, and the fourth has $(X, Y) = (-s^2t/v, 1)$. Hence every terminal generic point reaches the torus in at most four regular steps. The Laurent integral is regular there, and invariance pulls this regularity back. The torus, the eight boundary divisors, and these four terminal divisors exhaust the possible divisorial poles. Normality now gives the asserted pole divisor.

Let $Z_0 = (I_{r,s})_0$. It has no component in D , and $Z_0 \sim rD$, so $D \cdot Z_0 = 0$. Distinct effective curves on a smooth surface have positive local intersection at each meeting point. Thus $D \cap Z_0$ is empty. The zero divisors of the sections $1, I_{r,s} \in H^0(S, L_r)$ are rD, Z_0 ; they have no common point. The pole proves their independence. The denominator section is exactly 1, so the resulting morphism has the stated coordinate and the stated scheme fiber at infinity. $\square$

2.2 Integral node frames and genuine constants

The general relation between a torsion normal bundle and a Halphen pencil is classical [4, Proposition 2.2]. For the obstruction calculation we need the actual frames over R , including their compatibility with nonreduced base change.

Lemma 2.3. *For every commutative R -algebra B , the natural constant map gives $R\Gamma(S_B, \mathcal{O}_{S_B}) \simeq B[0]$. In fixed integral node frames the boundary complex is*

$$R\Gamma(D_B, \mathcal{N}_B^j) \simeq [B \xrightarrow{1-q_B^j} B], \quad j \geq 1, \quad (2.4)$$

in degrees zero and one. The frame changes and the degree-one scaling used in this description are fixed over R before any base change. Thus the complexes are compatible with reduction and with multiplication by a base parameter. On a field with $q = s$ of exact order r , $\mathcal{N}$ has exact order r and $h^0(D, \mathcal{N}^j) = 1$ precisely when $r \mid j$, and is zero otherwise.

Proof. First consider the blowup of $\mathbb{A}_B^2$ in its zero section. It is the total space of $\mathcal{O}_{\mathbb{P}_B^1}(-1)$, whose affine projection to $\mathbb{P}_B^1$ pushes its structure sheaf to $\bigoplus_{d \geq 0} \mathcal{O}(d)$. The two standard affine charts compute the higher cohomology of each summand as zero and the degree-zero cohomology of the total space as $B[a, b]$. Their finite Čech complex commutes with this direct sum. Thus a blowup b satisfies $b_*\mathcal{O} = \mathcal{O}$ and $R^i b_*\mathcal{O} = 0$ for $i > 0$. Coordinate changes and localization give the same calculation at every center of our surface; off the center the morphism is an isomorphism. Eight applications of Leray, followed by the four standard charts on $\mathbb{P}_B^1 \times_B \mathbb{P}_B^1$, give $R\Gamma(S_B, \mathcal{O}) = B[0]$. The map in degree zero is the actual constant inclusion throughout, not an unspecified free summand.

To determine the node gluing, let S' be the surface after just the four node blowups, with boundary D' . The last four blowups are at points p_i of D' , and $D \rightarrow D'$ is an isomorphism. If $\mathcal{L} = \mathcal{O}_{D'}(D')$, the divisor pullback gives

$$\mathcal{N} \simeq \mathcal{L}(-p_1 - p_2 - p_3 - p_4).$$

In the ordering (2.2), $\mathcal{L}$ has degree one on components 3, 5, 6, 8 and degree zero on the other four. At the successive nodes use the monomials

$$x^{-1}, x^{-1}, x/y, x/y, x, y, y, x^{-1}. \quad (2.5)$$

Each has valuation -1 on both adjacent boundary components. The adjacent order vectors displayed in the proof of Proposition 2.1 have determinant one, so their local characters give relative coordinates a, b at the node. The corresponding monomial in (2.5) is a unit times $(ab)^{-1}$, an actual local frame of $\mathcal{O}_{S'}(D')$. The points p_i are away from the nodes, so the same node frames apply after the four twists.

The zero-degree components have equal entering and leaving frames, giving propagation ratio one. On the remaining components a section before the twist is, successively,

$$ax^{-1} + bx/y, \quad bx/y + dx, \quad dx + ey, \quad ey + a'x^{-1}.$$

Vanishing at $x^2/y = \tau$, $y = 1$, $y/x = q$, and $xy = \tau$ gives

$$b/a = -\tau^{-1}, \quad d/b = -1, \quad e/d = -q^{-1}, \quad a'/e = -\tau.$$

The last component is traversed from $xy = \infty$ to $xy = 0$; this explains its direction in the last ratio. These sections vanish exactly at the prescribed points and become nowhere-zero frames of the twisted degree-zero bundles. Their full product is q^{-1} . Tensoring the same frames j times gives product q^{-j} ; no division by j or by $1 - q^j$ occurs.

The normalization sequence at a relative node is the exact difference sequence for $B[a, b]/(ab)$. Tensoring it by $\mathcal{N}^j$ and using the eight trivialized components gives a complex $[B^8 \rightarrow B^8]$, with node differences twisted by the above units. Eliminate the first seven differences by unit row and column operations fixed over R . The remaining difference is $1 - q^{-j}$. Multiplying its degree-one target by the unit $-q^j$ gives (2.4). Because every operation was integral and invertible, the same description works for arbitrary, including nonreduced, B and commutes with the indicated base maps.

Over a field a degree-zero section on any component is either zero or nowhere zero. Node compatibility propagates a nonzero section around the entire cycle, and it closes if and only if $s^{-j} = 1$. When it closes it trivializes $\mathcal{N}^j$, uniquely up to a constant; otherwise no nonzero section exists. This proves both the exact order and the section formula. In particular the same complexes apply to $B = \mathcal{O}_1/(\pi_1^2)$ and to k , with their original constant 1, and commute with $0 \rightarrow k \xrightarrow{\pi_1} \mathcal{O}_1/(\pi_1^2) \rightarrow k \rightarrow 0$. When $p = 2$ this first quotient has characteristic four; nothing in the chart calculation replaces it by a characteristic-two dual-number ring. $\square$

2.3 Completeness, connectedness, and generic smoothness

Theorem 2.4. *For s of exact order r and every $t \neq 0$,*

$$h^0(S, L_j) = 1 \quad (0 \leq j < r), \quad H^0(S, L_r) = k\langle 1, I_{r,s} \rangle.$$

The morphism $f = I_{r,s}$ is the complete primitive pencil, $f_\mathcal{O}_S = \mathcal{O}_{\mathbb{P}^1}$, and all its geometric fibers are connected. Its geometric generic fiber is smooth, integral, and of genus one, in every allowed characteristic. All finite fibers are complete curves contained in $\mathcal{U}$. Moreover $k(I_{r,s})$ is relatively algebraically closed in $k(S)$, and $k(S)/k(I_{r,s})$ is separably generated.*

Proof. Work first over an algebraically closed field. The constants calculation gives $H^1(S, \mathcal{O}) = H^2(S, \mathcal{O}) = 0$. Surface Riemann–Roch and $K_S = -D$, $D^2 = 0$ give $\chi(L_j) = 1$. Serre duality gives $h^2(L_j) = h^0(\mathcal{O}(-(j+1)D)) = 0$: an effective representative of the latter class would have negative intersection with an ample divisor. Thus $h^1(L_j) = h^0(L_j) - 1$. The exact sequences

$$0 \longrightarrow L_{j-1} \longrightarrow L_j \longrightarrow \mathcal{N}^j \longrightarrow 0 \tag{2.6}$$

and Lemma 2.3 give $h^0(L_j) = 1$ for $j < r$. At $j = r$, $H^1(L_{r-1}) = 0$ and $h^0(\mathcal{N}^r) = 1$, so $h^0(L_r) = 2$. This includes $r = 1$, when $L_{r-1} = \mathcal{O}$. Lemma 2.2 identifies the basis and proves that the complete system has no base point.

Take the Stein factorization $S \xrightarrow{h} B \xrightarrow{g} \mathbb{P}^1$ [15, Tag 03H0]. The curve B is normal, projective, and integral, hence smooth over our algebraically closed field, and $h_*\mathcal{O}_S = \mathcal{O}_B$. Leray injects $H^1(B, \mathcal{O}_B)$ into $H^1(S, \mathcal{O}_S) = 0$, so $B \simeq \mathbb{P}^1$. If $e = \deg g$, including its inseparable degree, the projection formula yields

$$2 = h^0(S, f^*\mathcal{O}(1)) = h^0(B, g^*\mathcal{O}(1)) = e + 1.$$

Thus $e = 1$, and g is an isomorphism. The geometric connectedness in Stein factorization gives connectedness of every geometric fiber, not merely of a generic point of a fiber.

We next verify generic smoothness without excluding characteristics two and three. A dominant morphism from an integral smooth surface to a smooth curve is flat: over every base DVR its local rings are torsion-free modules. Its fibers are Cartier divisors of pure dimension one. The relative Jacobian criterion therefore identifies the nonsmooth locus with the zero locus of the map from the pulled-back cotangent line of the base to $\Omega_{S/k}^1$. Let z be a parameter at infinity. Since the fiber is exactly rD , at a smooth point of D one has $f^*z = u^r a$, where a is a unit. With regular coordinates u, v ,

$$d(f^*z) = u^{r-1} \{ (ra + ua_u) du + ua_v dv \}.$$

The coefficient $ra + ua_u$ is a unit near $u = 0$, since r is invertible. At a node, $f^*z = u^r v^r a$ and

$$d(f^*z) = u^{r-1} v^{r-1} \{ v(ra + ua_u) du + u(ra + va_v) dv \}.$$

Both parenthesized factors are units near the node. These formulas give an open neighborhood of D in which all nonsmooth points lie in D . If a component of the closed nonsmooth locus dominated the base, properness would make it meet D . Its intersection with that neighborhood would then be a nonempty open subset contained in D , forcing the whole component into D , a contradiction. Hence all nonsmooth components are vertical. There are finitely many, and their proper images are finitely many base points. Removing these points gives a nonempty smooth locus on the base.

The smooth geometric fibers there are connected, so their distinct irreducible components cannot occur: components of a smooth curve are disjoint. They are consequently geometrically integral. For any fiber divisor $G \sim rD$, adjunction computes the integer

$$p_a(G) = 1 + \frac{G \cdot (G + K_S)}{2} = 1.$$

There is no division by the characteristic in this intersection formula. The smooth fibers thus have genus one. These statements also show that the geometric generic fiber is integral and smooth; its function field has no new algebraic constants over $k(c)$ and is separably generated. In particular $I_{r,s}$ cannot be a rational function of degree greater than one of another nonconstant rational function on S .

The complete support of the infinite fiber is D , so every finite fiber is disjoint from D and is still projective. Each terminal exceptional curve E satisfies $D \cdot E = 1$, hence $\deg(f|_E) = r > 0$. No finite-fiber component can be hidden entirely in a terminal curve. Thus the torus equation has the specified complete fiber as its closure, retaining every terminal point.

For an arbitrary original field, make the same constructions after extension of constants to an algebraic closure. The centers, sections, and parameter c are unchanged. Smoothness and geometric integrality are detected by faithful flat field extension, while proper cohomology commutes with a field extension. The computed section dimensions, $f_*\mathcal{O} = \mathcal{O}$, and all the geometric claims therefore hold over the original field as stated. $\square$

3 The actual spectral Jacobian

Fix a field k , units s, t with s of exact order r , and $K = k(c)$, where c is transcendental. In this section put $T = t^r$, $\varepsilon = (-1)^{r+1}$, and $n = 2r$. Let X/K be the generic fiber of the original pencil in Theorem 2.4. We will identify its Jacobian, including the torsor structure when $X(K)$ has no chosen point. General spectral-line-bundle and cyclic-Picard constructions provide the background [2, 9]. We give the rank-two construction and its inverse here, since the identity of the original state curve is essential.

3.1 A finite critical algebra before the spectral construction

Consider the projective cubic family $W \rightarrow \mathbb{A}_c^1$ with affine equation

$$F(u, v, c) = v^2 + cuv - \varepsilon T v - u^3 + T u^2 = 0, \quad O_W = [0 : 1 : 0]. \quad (3.1)$$

Its role at this point is purely algebraic; no identity with the original state curve is assumed.

Lemma 3.1. *The family W is projective and flat, its total space is geometrically regular, and each finite geometric fiber is integral and reduced. Its relative critical scheme is finite, with coordinate algebra*

$$\begin{aligned} \Gamma(Z_W, \mathcal{O}_{Z_W}) &= k[z]/((T - z^2)^2 - \varepsilon T z), \\ c &= \varepsilon - z - z^3/T. \end{aligned} \quad (3.2)$$

In particular W_K is geometrically smooth of genus one.

Proof. The homogeneous equation is $v^2 w + cuvw - \varepsilon T v w^2 - u^3 + T u^2 w = 0$. At infinity it has only O_W , where the derivative with respect to w is 1. Thus O_W is relatively smooth at every value of c . Every positive-degree component of a projective plane curve meets the line at infinity. For these cubics all such intersections would have to occur at O_W . Two components, or a repeated component, would make that point nonsmooth. Hence each geometric fiber has one component of multiplicity one. A plane hypersurface has no embedded components, so the fiber is integral and reduced. The cubic Hilbert polynomial is constant, giving flatness, and the projective equation gives properness. Each fiber has arithmetic genus one.

The relative derivatives and the derivative in the base direction are

$$F_u = cv - 3u^2 + 2Tu, \quad F_v = 2v + cu - \varepsilon T, \quad F_c = uv. \quad (3.3)$$

A singular point of a fiber cannot have $u = 0$: its possible values of v are $0, \varepsilon T$, and F_v is respectively $-\varepsilon T, \varepsilon T$. If $v = 0$, the remaining possibility is $u = T$, where $F_u = -T^2 \neq 0$. Thus every fiber singularity has $uv \neq 0$, and $F_c \neq 0$ there. The total space is smooth at those points by the absolute Jacobian criterion; it is smooth at all remaining points because it is relatively smooth there. These arguments also apply after an algebraic closure of k .

The relative differential presentation on the affine chart is

$$\Omega_{W/\mathbb{A}^1}^1 = (\mathcal{O}_W du \oplus \mathcal{O}_W dv)/(F_u du + F_v dv).$$

The critical scheme is therefore given by (F_u, F_v) , and it has no point at infinity. The preceding calculation places its entire support in $uv \neq 0$. This contains the scheme itself, including nilpotents: the closed complement would otherwise be a nonempty finite-type scheme and would have a geometric closed point. Eliminate c by $c = (\varepsilon T - 2v)/u$. Substitution into $F = 0$ and $uF_u = 0$ gives

$$v^2 = u^2(T - u), \quad \varepsilon T v = u^3.$$

Indeed the first substituted equation is $-v^2 - u^3 + T u^2 = 0$; the second, after using it, reduces to $\varepsilon T v - u^3 = 0$. Writing $z = v/u$ now gives

$$u = T - z^2, \quad v = z(T - z^2), \quad (T - z^2)^2 = \varepsilon T z.$$

Conversely, in the quotient by the displayed quartic, z is a unit because its constant term is T^2 . The equality $(T - z^2)^2 = \varepsilon T z$ makes $T - z^2$, and then v , units. The preceding formulas therefore define the inverse map and satisfy $F = F_u = F_v = 0$. They also give

$$c = T/z - 3z = \varepsilon - z - z^3/T,$$

where the last equality is the quartic relation divided by Tz . This proves (3.2) scheme-theoretically. The quartic is monic, so the algebra is finite-dimensional even when its roots repeat. Consequently its generic fiber over $k[c]$ is empty. The same calculation over an algebraic closure proves that W_K is geometrically smooth; its integral cubic fibers have arithmetic genus one. No step divided by two or three. $\square$

3.2 The two spectral charts and their four endpoints

Define E/K by gluing the two monic quadratic charts

$$\begin{aligned} \Phi(Z, \lambda) &= \lambda^2 - (T + cZ + Z^2)\lambda + \varepsilon Z^3 = 0, \\ \mu^2 - (1 + cV + TV^2)\mu + \varepsilon V &= 0, \quad V = Z^{-1}, \quad \mu = \lambda Z^{-2}. \end{aligned} \quad (3.4)$$

On $Z\lambda \neq 0$ the substitutions

$$u = \frac{\varepsilon TZ}{\lambda}, \quad v = \frac{\varepsilon T^2}{\lambda}, \quad Z = \frac{Tu}{v}, \quad \lambda = \frac{\varepsilon T^2}{v} \quad (3.5)$$

are mutually inverse, since

$$\frac{v^3}{T^3} \Phi(Tu/v, \varepsilon T^2/v) = -\varepsilon F(u, v, c).$$

They identify the locus $Z \neq 0, \infty$ with an open subset of the smooth W_K : here the spectral equation forces $\lambda \neq 0$. The two points over $Z = 0$ have $\lambda = 0, T$, with quadratic derivatives $-T, T$. The two points over $V = 0$ have $\mu = 0, 1$, with derivatives $-1, 1$. Thus every endpoint is smooth in every characteristic. This proves smoothness of the whole E , without a characteristic-dependent discriminant argument.

For the original spectral variable z define C/K by

$$\begin{aligned} \lambda^2 - (T + cz^r + z^{2r})\lambda + \varepsilon z^{3r} &= 0, \\ \mu^2 - (1 + cv^r + Tv^{2r})\mu + \varepsilon v^r &= 0, \quad v = z^{-1}, \quad \mu = \lambda z^{-2r}. \end{aligned} \quad (3.6)$$

Write P_0, P_T for $(z, \lambda) = (0, 0), (0, T)$, and Q_2, Q_1 for the points at $v = 0$ with $\mu = 1, 0$. The same derivatives show that z or v is a local parameter at each of these four points. Away from them, C is the pullback of E by $Z = z^r$, a finite étale map on the punctured line because rz^{r-1} is a unit. Consequently C is geometrically smooth as well.

The two-chart descriptions give finite flat degree-two maps to their respective projective lines. The bases $1, \lambda$ and $1, \mu$ have the transition prescribed in the equations, so

$$(\tau_C)_* \mathcal{O}_C = \mathcal{O} \oplus \mathcal{O}(-2r), \quad (\tau_E)_* \mathcal{O}_E = \mathcal{O} \oplus \mathcal{O}(-2). \quad (3.7)$$

These curves are projective and have $h^0(\mathcal{O}) = 1$ after algebraic closure. Smoothness and geometric connectedness imply geometric integrality, and their Euler characteristics yield $g(C) = 2r - 1$ and $g(E) = 1$. In particular, the genus of C is not the genus of the original state curve X .

Let $G = \langle \sigma \rangle$, with $\sigma(z, \lambda) = (sz, \lambda)$. On either monic quadratic chart every element is uniquely a sum of a polynomial and a polynomial times λ , or times μ at infinity. Taking invariants retains exactly the powers of z^r or v^r . Thus the quotient is the specified $\pi : C \rightarrow E$, $Z = z^r$, not merely a curve with the same function field. All four endpoints are fixed by the full group; their parameter maps are $z \mapsto z^r$ or $v \mapsto v^r$. For $r = 1$ the quotient is the identity. Finally, (3.5) extends uniquely to a smooth projective-curve isomorphism $E \simeq W_K$. At $O = (Z, \lambda) = (0, 0)$ the orders of Z, λ are $1, 3$, so u, v have pole orders $2, 3$; hence O maps to O_W . We use O as the origin on E .

Theorem 3.2. *The original generic curve X/K is an E -torsor over K , and $\text{Jac}(X) \simeq E$ over that same field. More precisely, the actual matrix $M_{r,s}$ defines a morphism*

$$\alpha : X \longrightarrow \text{Pic}^{2r}(C)$$

which identifies X with a geometric translate of $\pi^ \text{Pic}^0(E)$. The pullback $\pi^* : \text{Pic}^0(E) \rightarrow \text{Pic}^0(C)$ is a closed immersion with trivial scheme-theoretic kernel. No rational point on X is assumed.*

Proof. We separate the construction of the actual map from the group calculation which identifies its image.

The actual spectral module in families. On the nonempty torus open $X^\circ \subset X$, put $M(z) = A(s^{r-1}z) \cdots A(z)$ in the fixed gauge of (1.1). The original trace and determinant give

$$\det(\lambda I - M(z)) = \lambda^2 - (T + cz^r + z^{2r})\lambda + \varepsilon z^{3r}.$$

Let the generator λ of the spectral algebra act by $M(z)$ on $\mathcal{O}_{\mathbb{P}^1 \times X^\circ}^{\oplus 2}$. Cayley–Hamilton verifies its defining relation; at infinity use $v^n M(v^{-1})$. The finite-map equivalence between sheaves and modules defines a sheaf $\mathcal{L}$ on $C \times X^\circ$ with

$$(\tau_C)_* \mathcal{L} = \mathcal{O}^{\oplus 2}. \quad (3.8)$$

It is a line bundle. To check this also in the parameter direction, suppose that at a geometric parameter value and a finite $z = a$ the matrix were scalar, $M(a) = bI$. At (a, b) the adjugate of $\lambda I - M(z)$ is zero, so both partial derivatives of its determinant vanish. This would be a singular point of the already smooth C . At infinity the matrix is $P = \text{diag}(1, 0)$, also nonscalar. A nonscalar two-by-two matrix over a field has a cyclic vector: some vector e makes $e, M(a)e$ a basis. The same determinant remains a unit in a neighborhood in the family. Mapping the spectral-algebra basis $1, \lambda$ to $e, M(z)e$ there exhibits the module as free of rank one over that algebra. These neighborhoods prove the claimed line-bundle property of $\mathcal{L}$.

For every fiber L of this family, $\chi(L) = \chi(\mathcal{O}_{\mathbb{P}^1}^{\oplus 2}) = 2$. Since $g(C) = n - 1$, Riemann–Roch gives $\deg L = n$. The family therefore defines α on X° . Properness of the Picard component and the DVR valuative criterion extend it uniquely over every omitted point of the smooth projective curve X .

Recovery of the constant-conjugacy class. For a line bundle with pushdown (3.8), multiplication by the spectral coordinate is a map

$$\lambda : (\tau_C)_* L \longrightarrow (\tau_C)_* L(n).$$

A trivialization of the pushdown recovers a matrix polynomial of degree at most n . Two trivializations differ by a constant GL_2 matrix, not a rational z -dependent gauge. Conversely a line-bundle isomorphism pushes down to such a constant matrix intertwining the λ actions. This recovery works in families on the open locus where $H^0(C, L(-1)) = H^1(C, L(-1)) = 0$. There the pushdown is trivial, $H^0(C, L)$ has rank two, and its evaluation map identifies $H^0(C, L) \otimes \mathcal{O}_{\mathbb{P}^1}$ with the pushdown. These conditions hold on our image, because its twist pushes down to $\mathcal{O}(-1)^{\oplus 2}$. The rational point $P_0 \in C(K)$ allows rigidification of the Picard functor, so a universal line bundle and this family recovery are available on the relevant Picard open set. This is the particular spectral-correspondence argument needed here; it has not used a characteristic-zero version of a general moduli theorem.

A rational inverse to the original states. Write $M_j = [z^j]M$. The original matrix gives the three identities

$$M_n = P, \quad (M_{n-1})_{12} = 1, \quad M_0 = t^{r-1} A_0. \quad (3.9)$$

For the first, the highest coefficients multiply to $s^2 \sum_{i=0}^{r-1} i P^r = P$. For the second, a term of total degree $2r - 1$ contains one A_1 and otherwise P 's. Its $(1, 2)$ entry is zero unless A_1 is the rightmost factor, since a P on its right kills the second column. The remaining term has scalar multiplier $s^2 \sum_{i=1}^{r-1} i = 1$ and $(A_1)_{12} = 1$. Finally $\text{tr } A_0 = t$, $\det A_0 = 0$, so $A_0^2 = tA_0$ and $M_0 = A_0^r = t^{r-1}A_0$. These proofs include $r = 1$ with empty products interpreted as one. Since $(A_0)_{12} = -x$ and $(A_0)_{22} = x(y - 1)$, recovery is

$$x = -\frac{(M_0)_{12}}{t^{r-1}(M_{n-1})_{12}}, \quad y = 1 + \frac{(M_0)_{22}}{t^{r-1}x}. \quad (3.10)$$

The denominators are nonzero on X° . A constant conjugacy preserving $M_n = P$ is diagonal; it scales the two $(1, 2)$ entries together and preserves the $(2, 2)$ entry. Thus these formulas are independent of the remaining gauge.

They are rational functions on the Picard image, not just an algorithm on geometric points. Indeed on $H^0(C, L)$ the leading coefficient P splits the rank-two bundle into $\text{im } P \oplus \ker P$. The maps $PM_0(1 - P)$ and $PM_{n-1}(1 - P)$ between its two line bundles have scalar ratio $-t^{r-1}x$, and $\text{tr}((1 - P)M_0) = t^{r-1}x(y - 1)$. These are precisely the family-invariant ratios in (3.10). As x, y generate $K(X)$, they prove

$$K(\alpha(X)) \xrightarrow{\alpha^*} K(X) \quad \text{is an isomorphism.} \quad (3.11)$$

In particular α is nonconstant and birational onto its image, with total degree one. This excludes a hidden purely inseparable degree, which geometric generic injectivity alone would not exclude.

The fixed Picard difference. Cyclically rotating the original product gives $M(sz)A(z) = A(z)M(z)$, since $s^r = 1$. Hence $A(z)$ induces a rational homomorphism $a_L : L \dashrightarrow \sigma^*L$. It is invertible away from $z = 0, \infty$, because $\det A = z^3$. Its full cyclic composite is multiplication by the spectral coordinate:

$$(\sigma^{r-1})^* a_L \circ \cdots \circ \sigma^* a_L \circ a_L = M(z)|_L = \lambda. \quad (3.12)$$

Thus this map is not itself an ordinary G -linearization of L . The two spectral charts give

$$\text{div}_C(\lambda) = 3rP_0 - 2rQ_2 - rQ_1.$$

At P_0 , the other root is a unit tending to T , and the product of roots is εz^{3r} , giving order $3r$; at P_T the root is a unit. At Q_2 the coordinate μ tends to 1, giving pole order $2r$ for λ ; at Q_1 , μ has order r , giving pole order r . The divisor of a_L is supported at these four fixed points. Taking orders in (3.12) therefore gives the integer divisor identity

$$\text{div}(a_L) = D_A := 3P_0 - 2Q_2 - Q_1, \quad \sigma^*L \otimes L^{-1} \simeq \mathcal{O}_C(D_A). \quad (3.13)$$

This is independent of the state. By the extension of α and separation of the Picard functor it holds for every point of X , not just X° . Consequently, for any geometric base point P_* , all differences $\alpha(P) \otimes \alpha(P_*)^{-1}$ lie in

$$B_C = \ker(\sigma^* - 1 : \text{Pic}^0(C) \rightarrow \text{Pic}^0(C)).$$

The invariant component and the absence of a kernel. Write $J_E = \text{Pic}^0(E)$ and $J_C = \text{Pic}^0(C)$. The norm gives $\text{Nm}_\pi \pi^* = [r]$. Thus the kernel of $\pi^* : J_E \rightarrow J_C$ is a subgroup scheme of the finite étale group $J_E[r]$, since r is invertible. It suffices to exclude nontrivial geometric points. Over an algebraic closure of K , suppose $N \in J_E$ has trivial pullback. The pullback carries its natural G -action. In a trivialization, this action differs from function pullback by constants, since the global

units on the projective integral curve C are constant. On the fiber at the full-group fixed point P_0 , the natural action is the identity. Those constants must all be one. The resulting nowhere-zero section is G -invariant, and the quotient and projection formula give

$$H^0(C, \pi^* N)^G = H^0(E, N).$$

It descends to a nowhere-zero section of N : a zero would pull back to a zero. Thus N is trivial. The whole scheme-theoretic kernel is trivial, and the proper group homomorphism π^* is a closed immersion. Let $H_C = \pi^* J_E$.

The inclusion $H_C \subset B_C$ has dimension one. Finite-map cohomology and the averaging projector $r^{-1} \sum_{g \in G} g$ give

$$\mathrm{Lie}(B_C) = H^1(C, \mathcal{O}_C)^G = H^1(E, \mathcal{O}_E), \quad \dim \mathrm{Lie}(B_C) = 1.$$

Averaging is an exact direct-summand projector, so taking these invariants commutes with cohomology. The local dimension of B_C at the identity is at least one because it contains H_C , while its tangent dimension is one. Its local ring is therefore regular; translation gives smoothness of its identity component. The closed connected subgroup H_C of the same dimension is that component:

$$B_C^0 = H_C \simeq J_E. \quad (3.14)$$

For $r = 1$ the reasoning reduces to $C = E$ and $B_C = J_C$.

The full coset and descent over the original field. After choosing P_* geometrically, the difference morphism from X has connected image containing the identity, so it lies in $B_C^0 = H_C$. It is nonconstant by (3.11). Its projective image is therefore the whole one-dimensional group H_C . Hence $\alpha(X)$ is a complete geometric translate of H_C . It is smooth, and (3.11) makes the map birational to it. A birational morphism between smooth projective curves is an isomorphism, proving the asserted image description without a residual isogeny.

All the original surfaces, matrices, two spectral charts, and morphisms are defined over $K = k(c)$. If k is not algebraically closed, their geometric prerequisites are checked after the faithfully flat field extension $k(c) \subset \bar{k}(c)$ by the same equations and Theorem 2.4. No separability of this constant-field extension is required. Smoothness, geometric integrality, and the genus calculations descend; thus the module construction and the rational inverse were indeed over K . Let $Y = \alpha(X)$ with its K -structure. The K -defined action of J_E on $\mathrm{Pic}^n(C)$ by tensoring with $\pi^* N$ preserves Y , as can be checked after faithful flat geometric extension where it is a coset. The action map

$$J_E \times Y \longrightarrow Y \times Y, \quad (N, L) \longmapsto (L, L \otimes \pi^* N)$$

is an isomorphism after that extension, hence is an isomorphism over K . The closed immersion $\alpha : X \rightarrow Y$ is likewise an isomorphism over K . This gives X its J_E -torsor structure, and the rational origin $O \in E(K)$ identifies E with J_E . The torsor group of a genus-one curve is its Jacobian: with a geometric base point the identification is $P \mapsto \mathcal{O}_X(P - P_*)$, and changing the base point changes only a translation, not this group identification. It therefore descends to K , giving $\mathrm{Jac}(X) \simeq E$. The argument does not descend a geometric choice of P_* and does not assert that the torsor is trivial over K . $\square$

4 Complete smooth fibers and the Hasse polynomial

Return to the residue parameters of the cyclotomic setup. Let k be a perfect field of characteristic $p > 0$, let $\eta, t_0 \in k^\times$ with η of exact order m and $p \nmid m$, and put

$$J = I_{m, \eta}(x, y; t_0), \quad T = t_0^m, \quad \varepsilon = \varepsilon_m = (-1)^{m+1}.$$

For a finite closed energy, pass to its finite residue field extension l/k and write that energy as $h \in l$. The hypothesis throughout this section is that the *complete scheme fiber* $X_h = (J = h)$ of the original pencil is smooth over l . Smoothness only of its torus open set or of its reduced support would not be enough.

Theorem 4.1. *The curve X_h is smooth, projective, geometrically integral, and of genus one. It lies completely in the original open surface, including its intersections with the four terminal lines. Over an algebraic closure $\bar{l}$ it is isomorphic, at the same energy $c = h$, to the complete cubic W_h of (3.1). In particular this cubic and the complete spectral quotient E_h of (3.4) are smooth. No point or isomorphism is asserted to descend from $\bar{l}$ to l .*

Proof. By Theorem 2.4, the original morphism remains a surjective complete pencil after extension to $\bar{l}$, with geometrically connected fibers and infinite fiber exactly mD . Thus $X_{h,\bar{l}}$ is nonempty and connected. It is smooth by hypothesis, hence reduced. Two irreducible components of a smooth curve cannot meet, since a regular local ring is a domain. Its finitely many components would consequently be disjoint open and closed subsets. Connectedness implies that there is just one, proving geometric integrality. The curve is a closed fiber of the projective surface, is disjoint from D , and has divisor class mD . Adjunction gives

$$p_a(X_{h,\bar{l}}) = 1 + \frac{mD \cdot (mD + K_S)}{2} = 1,$$

which equals its geometric genus because it is smooth and integral. This proves the first statements for this selected fiber without using a classification of other finite fibers.

To compare complete models, set

$$A_0 = \bar{l}[c]_{(c-h)}, \quad A = A_0^h, \quad L = \text{Frac}(A), \quad \varpi = c - h. \quad (4.1)$$

Here A_0^h is the ordinary henselization. It is a DVR with the same residue field $\bar{l}$ and the same uniformizer ϖ . This operation is in the geometric residue problem at the fixed energy; it is not a ramified lift of an arithmetic state. Let

$$\mathcal{X} = S_{\bar{l}} \times_{\mathbb{P}_c^1} \text{Spec } A, \quad \mathcal{W} = W_{\bar{l}} \times_{\mathbb{A}_c^1} \text{Spec } A.$$

The first is proper and of finite presentation. It is flat because a nonzero base parameter is a non-zero-divisor in the local rings of the integral surface, and torsion-free modules over a DVR are flat. Localizing the smooth surface and then making this ind-étale henselian base change preserves regularity. Its generic fiber is smooth, projective, and geometrically integral of genus one by Theorem 2.4, and its special fiber is the geometrically integral reduced curve just proved. For $\mathcal{W}$, projectivity, flatness, regularity, geometrically integral reduced special fiber, and geometrically smooth genus-one generic fiber are exactly Lemma 3.1, preserved by this localization and henselization. Its point O_W gives a section. In particular, regularity here is a property of these specified base changes, not of arbitrary base change of a regular scheme.

Choose $q_0 \in X_{h,\bar{l}}(\bar{l})$. The morphism $\mathcal{X}/A$ is smooth in a neighborhood of q_0 , by flatness, finite presentation, and smoothness of that fiber at q_0 . An étale coordinate to $\mathbb{A}_A^1$ exists there; translate its residue value to zero. Pulling back the zero section gives an étale A -scheme with the specified residue point. Henselian lifting then supplies a section e_A through q_0 [15, Tag 04GG, Lemma 10.153.3(8)]. The section is only local over A ; this procedure makes no assertion that the original curve over $l(c)$ has a point.

Theorem 3.2 gives the actual E_L -torsor action on X_L , and (3.5) gives the specified origin-preserving isomorphism $\iota_L : E_L \simeq W_L$. The map $N \mapsto e_L + N$ trivializes this local torsor.

Composing its inverse with ι_L gives the unique equivariant generic isomorphism $\phi_L : X_L \rightarrow W_L$ satisfying $\phi_L(e_L) = O_W$.

Both models are relatively minimal regular proper models. Indeed in either one the special fiber F is the only vertical integral component, has multiplicity one, and is the principal Cartier divisor $\text{div}(\varpi)$. Its normal line bundle is therefore trivial, so $F^2 = \deg_F \mathcal{O}_F(F) = 0$. A curve of the first kind that could be contracted over the DVR would have to be a vertical component with negative normal degree. The unique available component has degree zero, excluding such a curve. This is minimality of a regular proper model in the sense of [15, Definition 55.8.4, Tag 0C2V], not merely minimality of a Weierstrass equation.

The positive-genus uniqueness and mapping theorem for minimal regular models applies: the identified generic curve is smooth, projective, geometrically integral, has $H^0(\mathcal{O}) = L$, and has genus one; both models are regular, proper, flat, and minimal. It extends the specified ϕ_L to an A -morphism ϕ_A . Apply the same theorem to ϕ_L^{-1} to obtain its extension in the other direction [15, Lemmas 55.10.1–55.10.2, Tags 0C6B and 0C9Z]. The two composites are the identity on the generic fiber. Flatness makes that fiber scheme-theoretically dense, and the targets are separated; hence both composites are the identity everywhere. This proves the full model isomorphism

$$\phi_A : \mathcal{X} \xrightarrow{\sim} \mathcal{W}, \quad \phi_A(e_A) = O_W. \quad (4.2)$$

It is A -linear, so in particular it fixes c and induces the claimed complete isomorphism at $c = h$. All terminal-fiber points are retained. Different choices of e_A need not give the same isomorphism.

The special cubic is now known to be smooth. Use the two original spectral charts with $c = h$. On $Z \neq 0, \infty$, (3.5) identifies them with a smooth open of W_h . At $Z = 0$ and infinity the four derivatives are again $-T, T, -1, 1$, so these endpoints are also smooth. The finite pushdown in (3.7) with degree two shows that this complete E_h is projective and geometrically connected of genus one; thus it is geometrically integral. The open isomorphism uniquely extends between the smooth projective curves to $E_{h,\bar{l}} \simeq W_{h,\bar{l}}$. Its geometric smoothness is detected over $\bar{l}/l$, completing the assertion about E_h as well. $\square$

Proposition 4.2. *On the complete smooth spectral curve E_h the specified form*

$$\omega = \frac{dZ}{2\lambda - (T + hZ + Z^2)} \quad (4.3)$$

is a nonzero regular differential. With the p^{-1} -linear Cartier convention its value is

$$\begin{aligned} \mathcal{C}(\omega) &= H_p(T, h; \varepsilon)^{1/p} \omega, \\ H_p(T, h; \varepsilon) &= \begin{cases} [Z^{p-1}]((T + hZ + Z^2)^2 - 4\varepsilon Z^3)^{(p-1)/2}, & p \neq 2, \\ h, & p = 2. \end{cases} \end{aligned} \quad (4.4)$$

The polynomial H_p is monic of degree $p - 1$ in h . For the original complete smooth curve X_h one has

$$\text{Jac}(X_h) \text{ is geometrically supersingular} \iff H_p(T, h; \varepsilon) = 0 \text{ in } l.$$

No simplicity of a root of H_p is assumed.

Proof. Write $\Phi = \lambda^2 - (T + hZ + Z^2)\lambda + \varepsilon Z^3$. At a finite point where $\Phi_\lambda \neq 0$, the expression dZ/Φ_λ is a regular local generator of the cotangent line. Where that derivative vanishes, smoothness makes Φ_Z a unit and the equation $d\Phi = 0$ gives the equivalent expression $-d\lambda/\Phi_Z$. These expressions

agree on their overlap and include every finite branch point. On the infinity chart $V = Z^{-1}$, $\mu = \lambda Z^{-2}$ the form is

$$-\frac{dV}{2\mu - (1 + hV + TV^2)}.$$

At both endpoints $V = 0$ the denominator is a unit. This verifies regularity on the entire complete curve, and any one of these local expressions proves that it is nonzero. Since the curve has genus one, a nonzero regular differential has an effective zero divisor of degree $2g - 2 = 0$, so is nowhere zero.

We use the convention $\mathcal{C}(g^p\nu) = g\mathcal{C}(\nu)$ and

$$\mathcal{C}\left(\sum_j a_j Z^j dZ\right) = \sum_{i \geq 0} a_{pi+p-1}^{1/p} Z^i dZ.$$

These conventions distinguish the Hasse coefficient from its p th root in the Cartier–Manin matrix; the distinction and semilinear convention are discussed in the corrected version of [1, Section 2.5]. The projection variable Z is separating: its quadratic equation has nonzero derivative in λ as a rational function, also in characteristic two. Thus the displayed Cartier rules apply in the curve's function field.

For odd p set $Y = 2\lambda - (T + hZ + Z^2)$. Then

$$Y^2 = (T + hZ + Z^2)^2 - 4\varepsilon Z^3, \quad \omega = Y^{-p}(Y^{p-1}dZ).$$

The polynomial Y^{p-1} in Z has degree $2p - 2$. Among its exponents the only one congruent to $p - 1$ modulo p is $p - 1$. The Cartier rules therefore give

$$\mathcal{C}(\omega) = Y^{-1}\mathcal{C}(Y^{p-1}dZ) = H_p(T, h; \varepsilon)^{1/p} \frac{dZ}{Y}.$$

For $p = 2$, put $S_0(Z) = T + hZ + Z^2$. Formula (4.3) is $dZ/S_0(Z)$, so

$$\mathcal{C}(\omega) = S_0(Z)^{-1}\mathcal{C}(S_0(Z)dZ) = h^{1/2} \frac{dZ}{S_0(Z)}.$$

The constant and quadratic terms contribute nothing and the linear term contributes $h^{1/2}dZ$. These are identities of regular differentials on the complete curve, by the regularity just proved. For odd p , the term of degree $p - 1$ in h in the coefficient formula can only come from $(h^2Z^2)^{(p-1)/2}$, and has coefficient one. Terms of smaller h -degree cannot change it. For $p = 2$ monicity follows from $H_2 = h$.

Pull the specified differential back along the two complete geometric isomorphisms in Theorem 4.1. Cartier commutes with pullback by curve isomorphisms, so the pulled back form on $X_{h,\bar{l}}$ has the same displayed coefficient. The Cartier operator on a genus-one curve is zero precisely when its geometric Jacobian is supersingular. This criterion also applies when $X_h(l)$ has no chosen point, by taking the Jacobian after geometric base extension. A different isomorphism or differential basis multiplies the form by a nonzero scalar a ; $\mathcal{C}(a\omega) = a^{1/p}\mathcal{C}(\omega)$ shows that this cannot change whether the operator is zero. Finally $H_p(T, h; \varepsilon) \in l$ and $l \hookrightarrow \bar{l}$ is injective. Its geometric zero condition is exactly its zero condition in l . Only that condition is being returned to the original field, not a chosen geometric point, a canonical isomorphism, or an arbitrarily normalized differential. $\square$

Remark 4.3. The smooth-fiber hypothesis also supplies the two differential directions used later. Along X_h , the map J is smooth, so

$$0 \longrightarrow \mathcal{O}_{X_h} dJ \longrightarrow \Omega_{\mathcal{U}_l/l}^1|_{X_h} \longrightarrow \Omega_{X_h/l}^1 \longrightarrow 0$$

is exact with locally free terms. A nonzero regular tangent differential is nowhere zero because X_h has genus one. Consequently a lift of such a differential and the normal direction dJ give a local basis at every point of the complete fiber, including the four terminal charts. This conclusion does not concern only the torus and makes no assertion about a singular fiber.

5 Integral trace calculus and global divided forms

The division defining α_{mp^a} must take place before reduction. We establish this division on the original matrix product, extend the resulting form to the full open surface, and compute its residue. These calculations also fix the trace identities used in the first-order computations below. Throughout this section, d differentiates the two state variables and fixes the base ring, the unit time, and the spectral variable.

5.1 Cyclic insertion before reduction

Write $N = p^a$, $r = mN$, $s = s_a$, and $t = t_a$. On the original torus let

$$R = \mathcal{O}_a[x^{\pm 1}, y^{\pm 1}], \quad \Omega = R dx \oplus R dy.$$

The entries of the matrix in (1.1) lie in $R[z]$. The trace convention in (1.2) gives $I_{r,s} = [z^r] \operatorname{tr} M_{r,s}(z)$.

Lemma 5.1 (Integral cyclic insertion). *Set*

$$Q_r(z) = \operatorname{tr}(dA(z)A(s^{r-1}z) \cdots A(sz)).$$

Then, as an equality in Ω before reduction,

$$dI_{r,s} = r[z^r]Q_r(z), \quad \alpha_r = m[z^r]Q_r(z) \in \Omega. \quad (5.1)$$

Proof. Differentiate the descending product $M_{r,s}$. Its j th summand has $dA(s^j z)$ in place of $A(s^j z)$. Cyclically rotate its trace so that this factor is first, and put $w = s^j z$. The remaining factors, in their order, are $A(s^{r-1}w), \dots, A(sw)$: indices are reduced modulo r , using $s^r = 1$. No two matrix factors have been interchanged. The summand is therefore $Q_r(s^j z)$. Its coefficient of z^r is $[z^r]Q_r(z)$, because $s^{jr} = 1$. Summing the r insertions proves the first equality. Since $r = mp^a$, the second equality follows in characteristic zero and exhibits its integral coefficients. Specializing the unit t does not change the identity. $\square$

5.2 The rank-two recurrence and its coefficient

Let k be the residue field, $\eta = \bar{s}$, and

$$B(z) = A(\eta^{m-1}z; \bar{t}) \cdots A(z; \bar{t}), \quad J = I_{m,\eta}, \quad T = \bar{t}^m, \quad \varepsilon = (-1)^{m+1}.$$

The residue of $M_{r,s}$ is B^N . For a matrix over a commutative ring of characteristic p , $\operatorname{tr}(B^{p^a}) = (\operatorname{tr} B)^{p^a}$. Indeed, the assertion holds for the two eigenvalues of the generic matrix over its splitting

field, hence is a polynomial identity in its entries. Consequently $\bar{I}_{r,s} = J^N$. Differentiating this last equality would lose the information retained by Lemma 5.1.

Put $A_j = A(\eta^j z; \bar{t})$ and

$$C_j = A_{m-1} \cdots A_{j+1} dA_j A_{j-1} \cdots A_0, \quad dB = \sum_{j=0}^{m-1} C_j.$$

Empty products are identity matrices. Reducing the integral insertion formula gives

$$\bar{\alpha}_r = m[z^{mN}] \operatorname{tr}(B^{N-1} C_0).$$

The same cyclic rotation used above shows that the coefficient of z^{mN} in $\operatorname{tr}(B^{N-1} C_j)$ is independent of j : the spectral rescaling contributes $\eta^{jmN} = 1$. Summing the m terms, rather than cancelling the vanishing integer N , yields

$$\bar{\alpha}_r = [z^{mN}] \operatorname{tr}(B^{N-1} dB). \quad (5.2)$$

Define integer polynomials by

$$U_{-1} = 0, \quad U_0 = 1, \quad U_n(S, D) = SU_{n-1}(S, D) - DU_{n-2}(S, D) \quad (n \geq 1). \quad (5.3)$$

For any rank-two matrix and any derivation of its coefficient ring,

$$\operatorname{tr}(B^{n-1} dB) = U_{n-1}(S, D) dS - U_{n-2}(S, D) dD, \quad S = \operatorname{tr} B, \quad D = \det B, \quad n \geq 2. \quad (5.4)$$

For $n = 2$, the identity follows from $dD = \operatorname{tr}((S \operatorname{id} - B)dB)$. For larger n , Cayley–Hamilton gives $B^{n-1} = U_{n-2}B - DU_{n-3} \operatorname{id}$; substitution of the $n = 2$ identity and the recurrence proves (5.4). This argument works over the integer coefficient ring, including repeated eigenvalues, and does not introduce a discriminant denominator.

For the original block, setting $Z = z^m$ gives

$$S = T + JZ + Z^2, \quad D = \varepsilon Z^3, \quad dS = Z dJ, \quad dD = 0.$$

It follows from (5.2) that

$$\bar{\alpha}_r = [Z^{N-1}] U_{N-1}(T + JZ + Z^2, \varepsilon Z^3) dJ. \quad (5.5)$$

Lemma 5.2 (Residue multiplier). *With $H = H_p(T, J; \varepsilon)$ and $\sigma = (p^a - 1)/(p - 1)$ as in the main statements, the coefficient in (5.5) is H^σ . The polynomial $H_p(T, h; \varepsilon)$ is monic of degree $p - 1$ in h .*

Proof. The integer identity

$$U_{N-1}(u + v, uv) = \frac{u^N - v^N}{u - v}$$

is an identity of polynomials; the quotient denotes the polynomial sum, also when $u = v$. In characteristic p , it is $(u - v)^{N-1}$. The injective map from symmetric polynomials in S, D to polynomials in u, v therefore gives

$$U_{N-1}(S, D) = \begin{cases} (S^2 - 4D)^{(N-1)/2}, & p \neq 2, \\ S^{N-1}, & p = 2. \end{cases}$$

For odd p , set

$$f(Z) = ((T + hZ + Z^2)^2 - 4\varepsilon Z^3)^{(p-1)/2}.$$

It has degree at most $2p - 2$, and the required polynomial is $\prod_{i=0}^{a-1} f(Z)^{p^i}$. A term contributing to Z^{p^a-1} has indices n_i with $0 \leq n_i \leq 2p - 2$ and $\sum_i n_i p^i = p^a - 1$. Reduction modulo p forces $n_0 = p - 1$, the only integer in that interval congruent to $p - 1$. Subtracting this value and dividing by p repeats the argument for each subsequent index. Thus every $n_i = p - 1$, and the coefficient is

$$\prod_{i=0}^{a-1} ([Z^{p-1}]f(Z))^{p^i} = H_p(T, h; \varepsilon)^{1+p+\dots+p^{a-1}}.$$

For $p = 2$, use $f(Z) = T + hZ + Z^2$ instead. The only odd index in $[0, 2]$ is one, so the same argument gives h^{2^a-1} . Finally, in odd characteristic the term h^{p-1} in H_p can only come from $(h^2 Z^2)^{(p-1)/2}$ and has coefficient one; all other choices have lower h -degree. For $p = 2$, monicity follows from $H_2 = h$. $\square$

5.3 Extension to the full open surface

Proposition 5.3 (Regularity before reduction). *On the original open model $\mathcal{U}_a$, the form $\alpha_{mp^a} = p^{-a} dI_{mp^a, s_a}$ is regular and satisfies*

$$\bar{\alpha}_{mp^a} = H^\sigma dJ \quad \text{on all of } \mathcal{U}_k. \quad (5.6)$$

Both assertions include the four complete terminal affine lines.

Proof. The eight blowups of Proposition 2.1 are blowups of sections over $\mathcal{O}_a$. Their two local charts are relative affine planes, and the last centers have unit coordinates $1, t, t, s$ on distinct boundary components. Thus the surface is smooth over $\mathcal{O}_a$, its special fiber is geometrically integral, and its relative cotangent sheaf is locally free.

First regard $I_{r,s}$ as a rational section of $\mathcal{O}(rD)$. At every horizontal prime divisor it is regular by the polar-divisor calculation of Section 2, applied over the fraction field. At the unique vertical prime divisor its Laurent coefficients are integral, and that divisor's generic point lies in the original torus. On a normal Noetherian scheme a locally free module is the intersection of its localizations at prime divisors inside its generic fiber [15, Tag 0AVB]. Applying this fact to the line bundle extends $I_{r,s}$ to a regular section and hence to a regular function on $\mathcal{U}_a$.

The rational one-form $p^{-a} dI_{r,s}$ is regular at horizontal prime divisors, since p is invertible there. At the vertical generic point, Lemma 5.1 supplies integral coefficients. The same intersection property, now for the locally free cotangent sheaf, extends it uniquely across the whole open surface. On the special fiber, both sides of (5.6) are regular sections of a locally free sheaf. They agree on the dense torus by Lemma 5.2, so agree on the geometrically integral special fiber. No terminal line has been removed in this extension. $\square$

For later congruences the base need not be reduced. Density of the underlying point set is then insufficient, so we record the actual algebraic restriction property of the terminal charts.

Lemma 5.4 (Restriction on nonreduced terminal charts). *Let B be any quotient of a base ring used above. On a terminal chart, the coordinate ring is a localization of $B[u, v]$ by the displayed unit factors in (2.1). Localization at u is injective on this ring and on its free module with basis du, dv . Consequently an equality of regular forms on this chart can be tested after inverting u , also over a nonreduced B .*

Proof. Multiplication by u is injective in the polynomial ring $B[u, v]$, by comparison of its coefficients, regardless of nilpotents in B . It remains injective after localization by the unit factors of

the chart. Hence the map to the localization at u is injective. The same statement holds componentwise on the free relative cotangent module. These charts cover the terminal lines and meet the original torus where u is invertible. Applying the assertion to each chart proves the claim. $\square$

Taking the two coefficients of (5.6) in any local frame gives

$$\overline{\mathfrak{c}(\alpha_{mp^a})} = (H^\sigma) \mathfrak{c}(dJ).$$

On a smooth finite fiber, dJ is a local coordinate differential by Theorem 2.4. The displayed ideal is then the Hasse divisor with its multiplicities multiplied by σ ; it does not yet determine the unreduced critical ideal. Lemma 5.2 and the original Laurent leading term $J = -\bar{t}^m x^{-m} + O(x^{-m+1})$ also show that $H^\sigma dJ \neq 0$: $p \nmid m$ makes the leading coefficient of J_x nonzero. Thus the generic common uniformizer order of dI_{mp^a} on the special surface is exactly $a\varphi(p^a)$. Pointwise orders along lifted states require the vertical information computed next.

6 The complete first-layer critical ideal

Fix $a = 1$ and write $\mathcal{O} = \mathcal{O}_1$, $\pi = \pi_1$, $s = \tilde{\eta}\zeta_p$, $t = t_1$, $I = I_{mp,s}$, and $\alpha = p^{-1}dI$. The residue field k is perfect. We first compute the obstruction to lifting the original small-order integral J . An integer trace congruence then identifies that obstruction with the first tangent component of α on a supersingular fiber. The connecting homomorphisms are the usual ones; the issue is their value on this particular section and this particular trace.

6.1 The actual boundary Bockstein

Write S, D for the eight-center model and its boundary over $\mathcal{O}$. Let $A_2 = \mathcal{O}/(\pi^2)$, $L_n = \mathcal{O}_S(nD)$, and $\mathcal{N} = \mathcal{O}_D(D)$. Subscripts A_2 and k denote the actual base changes of these sheaves and of the eight-center surface. The coefficient sequence $0 \rightarrow k \xrightarrow{\pi} A_2 \rightarrow k \rightarrow 0$ defines a Bockstein

$$\beta_L : H^0(S_k, L_{m,k}) \longrightarrow H^1(S_k, L_{m,k}).$$

For $p = 2$, $\pi = -2$ and A_2 has characteristic four; it is not replaced by a characteristic-two dual-number ring.

Proposition 6.1 (The specified obstruction class). *The kernel of β_L is the actual constant line $k\langle 1 \rangle$. For every complete smooth finite fiber $X = (J = h)$, the original section 1 trivializes $L_m|_X$ and induces an isomorphism*

$$\rho_X : H^1(S_k, L_{m,k}) \xrightarrow{\sim} H^1(X, \mathcal{O}_X).$$

If j_i are local regular lifts of J , then

$$0 \neq \kappa_J := \rho_X \beta_L(J) = \left[\overline{(j_j - j_i)/\pi} \right]_X. \quad (6.1)$$

The assertion holds after the allowed perfect residue-field extensions needed to specify h .

Proof. The actual blowup-chart calculation of Lemma 2.3 gives $H^0(S_A, \mathcal{O}) = A\langle 1 \rangle$ and $H^{>0}(S_A, \mathcal{O}) = 0$ for $A = A_2, k$, with the same constant section and maps of coefficients. The node frames in (2.4), fixed first over $\mathcal{O}$, compute the boundary cohomology by

$$K_{j,A} = [A \xrightarrow{1-s_A^j} A] \quad \text{in degrees zero and one.} \quad (6.2)$$

In particular, the elimination of seven invertible node pairs and the degree-one scaling by $-s^j$ are the same before both base changes. They commute with reduction, multiplication by π , and connecting homomorphisms.

For $1 \leq j < m$, $1 - \eta^j \neq 0$, so the arrow in (6.2) is a unit over A_2 and k . The exact sequences

$$0 \longrightarrow L_{j-1,A} \longrightarrow L_{j,A} \longrightarrow \mathcal{N}_A^j \longrightarrow 0$$

therefore give, by induction, $H^0(S_A, L_{j,A}) = A\langle 1 \rangle$ and $H^{>0}(S_A, L_{j,A}) = 0$ for $0 \leq j < m$. If $m = 1$, the induction interval is empty and the assertion is precisely the constants calculation.

At $j = m$ over k , the boundary differential vanishes. Its two cohomology groups are k , and the same exact sequence gives

$$\begin{aligned} 0 \longrightarrow k\langle 1 \rangle \longrightarrow H^0(S_k, L_{m,k}) &\xrightarrow{r_0} H^0(D_k, \mathcal{N}_k^m) \longrightarrow 0, \\ r_1 : H^1(S_k, L_{m,k}) &\xrightarrow{\sim} H^1(D_k, \mathcal{N}_k^m). \end{aligned}$$

Naturality for restriction to the boundary gives $r_1\beta_L = \beta_D r_0$. In the fixed boundary coordinates,

$$1 - s^m = 1 - (1 + \pi)^m \equiv -m\pi \pmod{\pi^2}.$$

Lifting the degree-zero element 1, applying the differential, dividing by π , and reducing shows that β_D is multiplication by $-\bar{m}$. It is invertible since $p \nmid m$. Thus $\ker \beta_L = \ker r_0 = k\langle 1 \rangle$. The original Laurent leading term makes J nonconstant, so $\beta_L(J) \neq 0$. This computation does not identify $r_0(J)$ with an arbitrarily normalized boundary basis: if its coordinate is b_J , the image is exactly $-\bar{m}b_J$.

The complete finite fiber is disjoint from D_k . Its actual defining section yields

$$0 \longrightarrow \mathcal{O}_{S_k} \xrightarrow{J-h\cdot 1} L_{m,k} \longrightarrow L_{m,k}|_X \longrightarrow 0.$$

The section 1 has divisor mD_k , so its restriction to X is nowhere zero and supplies the stated trivialization. The vanishing of $H^1(S_k, \mathcal{O})$ and $H^2(S_k, \mathcal{O})$ proves that the actual restriction ρ_X is an isomorphism. Represent the local lifted sections using this same trivialization. With the Cech convention $c_j - c_i$, the Bockstein is represented by $(j_j - j_i)/\pi$; restriction gives (6.1). Changing lifts adds a coboundary. Hence the class is the specified restriction of the specified Bockstein and is nonzero on each X . $\square$

6.2 The original prime trace congruence

The next calculation keeps both the ramification and the nonresonant terms of the original m -block. Set

$$\begin{aligned} B_s(z) &= A(s^{m-1}z; t) \cdots A(z; t), \quad r_0 = s^m, \\ T_0 &= t^m, \quad C = s^{m(m-1)}, \quad \delta = s^{3m(m-1)/2}, \quad j_t = [z^m] \operatorname{tr} B_s(z), \\ \lambda &= 1 \ (p \neq 2), \quad \lambda = -1 \ (p = 2). \end{aligned}$$

Here r_0 is a primitive p th root, not the length mp . Define the integer trace polynomials

$$L_0(S, D) = 2, \quad L_1(S, D) = S, \quad L_n = SL_{n-1} - DL_{n-2},$$

and the polynomial with state-constant coefficients

$$F(v) = \lambda[Z^p]L_p(T_0 + vZ + CZ^2, \delta Z^3). \tag{6.3}$$

Lemma 6.2 (Prime trace congruence). *On the original torus,*

$$I - F(j_t) \in p\pi\mathcal{O}[x^{\pm 1}, y^{\pm 1}], \quad F(v) = \lambda v^p + pQ(v), \quad Q \in \mathcal{O}[v]. \quad (6.4)$$

Moreover, $F'(v)/p \in \mathcal{O}[v]$ and

$$\overline{F'(v)/p} = H_p(T, v; \varepsilon_m), \quad T = \bar{t}^m. \quad (6.5)$$

Proof. The constant matrix A_0 has trace t and determinant zero, hence $A_0^m = t^{m-1}A_0$. The highest matrix coefficient is $C \operatorname{diag}(1, 0)$, and multiplication of determinants gives $\det B_s = \delta z^{3m}$. Thus the constant and top trace coefficients are T_0, C . After reduction, cyclic rotation gives $\operatorname{tr} B_\eta(\eta z) = \operatorname{tr} B_\eta(z)$. Since η has exact order m , every nonresonant trace coefficient is divisible by π . Writing

$$S_*(Z) = T_0 + j_t Z + CZ^2, \quad E_s(z) = \operatorname{tr} B_s(z) - S_*(z^m),$$

we have $E_s \in \pi R[z]$ and $[z^{im}]E_s = 0$, where $R = \mathcal{O}[x^{\pm 1}, y^{\pm 1}]$.

Write $B_s(z) = \sum_{i=0}^{2m} B_i z^i$. For an exponent word $\mathbf{i} = (i_0, \dots, i_{p-1})$ of total degree mp , put

$$w(\mathbf{i}) = \sum_{\ell=0}^{p-1} \ell i_\ell, \quad \tau(\mathbf{i}) = \operatorname{tr}(B_{i_{p-1}} \cdots B_{i_0}).$$

The contribution to the original descending block product is $r_0^{w(\mathbf{i})} \tau(\mathbf{i})$. Under cyclic shift, τ is unchanged and the weight changes by $p(m - i_{p-1})$. Thus each cyclic orbit has constant phase. Its size is one or p . The unique size-one word has all indices m , with phase $r_0^{mp(p-1)/2} = \lambda$: the phase is one for odd p , and minus one for $p = 2$, when m is odd. Its contribution cancels against $\lambda[z^{mp}] \operatorname{tr} B_s(z)^p$. Every size- p orbit contributes

$$p(r_0^{w(\mathbf{i})} - \lambda) \tau(\mathbf{i}) \in p\pi R.$$

We have proved $I - \lambda[z^{mp}] \operatorname{tr} B_s^p \in p\pi R$ without commuting matrix factors.

Cayley–Hamilton gives $\operatorname{tr} B_s^p = L_p(\operatorname{tr} B_s, \det B_s)$. In symmetric variables $L_p(u+v, uv) = u^p + v^p$, so $L_p(S, D) = S^p + pV_p(S, D)$ for an integer polynomial V_p . Replacing $S_* + E_s$ by S_* in the pV_p part changes it by a multiple of $p\pi$. For the p th-power part, expand in the scalar trace coefficients. The only pure p th power contributing to degree mp has index m and is j_t^p on both sides. Every other multinomial coefficient is divisible by p . An uncanceled such term contains a nonresonant coefficient, hence a further factor π . This proves that the resonant coefficient changes by a multiple of $p\pi$, even when several nonresonant factors return to resonant degree. The first assertion of (6.4) follows.

The identity $L_p = S^p + pV_p$ also implies $F(v) \equiv \lambda v^p \pmod{p}$, proving the second assertion. For the derivative, use the companion matrix $M = \begin{pmatrix} S & -D \\ 1 & 0 \end{pmatrix}$. Cyclic insertion in its trace gives

$$\partial_S L_p = p(M^{p-1})_{11} = pU_{p-1}(S, D).$$

Consequently $F'(v)/p = \lambda[Z^{p-1}]U_{p-1}(T_0 + vZ + CZ^2, \delta Z^3)$. Here $\bar{C} = 1$, $\bar{\delta} = \varepsilon_m$, and $\bar{\lambda} = 1$. For the middle equality, if m is odd the exponent of η in δ is a multiple of m ; if m is even then p is odd and $\eta^{m/2} = -1$. Applying Lemma 5.2 at $a = 1$ now gives (6.5), including $p = 2$. $\square$

Lemma 6.3 (Ramified integer Taylor quotient). *The quotient*

$$\mathcal{T}_F(v, f) = \frac{F(v + \pi f) - F(v)}{p\pi}$$

is a polynomial in $\mathcal{O}[v, f]$, and

$$\overline{\mathcal{T}_F(v, f)} = H_p(T, \bar{v}; \varepsilon_m) \bar{f} - \bar{f}^p. \quad (6.6)$$

Proof. The cyclotomic equation $\Phi_p(1 + \pi) = 0$ yields

$$\rho := \frac{\pi^{p-1}}{p} = -1 - \sum_{j=2}^{p-1} \frac{\binom{p}{j}}{p} \pi^{j-1} \in \mathcal{O}^\times, \quad \bar{\rho} = -1.$$

Therefore $\overline{\lambda\rho} = -1$ for all p . At $p = 2$, the integer value is $\lambda\rho = 1$, whose residue is indeed -1 ; this is not the characteristic-zero equality $1 = -1$. Using $F = \lambda v^p + pQ$, the quotient is

$$\lambda v^{p-1} f + \lambda \sum_{j=2}^{p-1} \frac{\binom{p}{j}}{p} \pi^{j-1} v^{p-j} f^j + \lambda \rho f^p + \frac{Q(v + \pi f) - Q(v)}{\pi}.$$

Every term is integral. Reduction kills the middle sum and gives (6.6) by (6.5). The sum is empty for $p = 2$. The quotient is formed as a polynomial before any specialization; on a ring with torsion its value means this polynomial value, not a newly attempted division by a zero divisor. $\square$

For each local regular lift j_i of J , Lemmas 6.2 and 6.3 give $I - F(j_i) \in p\pi\mathcal{O}$ on its intersection with the torus. The functions I and $F(j_i)$ are regular on the full chart by Proposition 5.3. Apply Lemma 5.4 over $\mathcal{O}/(p\pi)$ to their difference. It follows that

$$G_i := \frac{I - F(j_i)}{p\pi} \quad \text{is regular on each full local chart.} \quad (6.7)$$

This proves the congruence at every terminal point. It does not assert that the torus function j_t has a global regular lift.

6.3 The image-sheaf class and the two coefficients

Proposition 6.4 (The actual first tangent class). *Suppose $X = (J = h)$ is a complete smooth finite fiber with $H_p(T, h; \varepsilon_m) = 0$. The local forms $d(\bar{G}_i|_X)$ glue to a regular differential ν on X , and*

$$\partial\nu = \text{Fr}_* \kappa_J \neq 0 \quad \text{in } H^1(X, \mathcal{O}_X^p). \quad (6.8)$$

Here ∂ belongs to $0 \rightarrow \mathcal{O}_X^p \rightarrow \mathcal{O}_X \xrightarrow{d} B_X^1 \rightarrow 0$, where $B_X^1 = \text{im } d$, and Fr maps the additive sheaf $\mathcal{O}_X$ to its image sheaf $\mathcal{O}_X^p$. The differential ν is nowhere zero on the complete curve.

Proof. Set $\hat{H}(v) = F'(v)/p$. Differentiating (6.7) before reduction gives the exact local identity

$$\alpha = \hat{H}(j_i) dj_i + \pi dG_i. \quad (6.9)$$

On an overlap write $j_j = j_i + \pi \tilde{f}_{ij}$ and $f_{ij} = \tilde{f}_{ij}|_X$. The sign convention fixed in Proposition 6.1 and the Taylor formula give

$$\bar{G}_j - \bar{G}_i = -H_p(T, J; \varepsilon_m) \tilde{f}_{ij} + \tilde{f}_{ij}^p.$$

Restriction to the Hasse-zero fiber therefore gives the exact equality $\bar{G}_j|_X - \bar{G}_i|_X = f_{ij}^p$. Its differential vanishes, proving that the local exact forms glue. Equation (6.9) identifies their common value with the tangent class of the original form after its normal term is subtracted and division by π is performed. Replacing j_i by another lift changes the restricted primitive by a p th power, so does not change ν .

We justify the kernel sheaf used in the connecting sequence. The function field K of a smooth curve over a perfect field has a separating transcendence element z . If $[K : k(z)] = n$, then

$[K^p : k(z^p)] = n$ and $[K : k(z^p)] = pn$, so $[K : K^p] = p$. Since $dz \neq 0$, the elements $1, z, \dots, z^{p-1}$ form a basis over K^p ; differentiation proves $\ker(d : K \rightarrow \Omega_{K/k}^1) = K^p$. If a regular function $g = b^p$ has zero differential, then at each local discrete valuation ring $pv(b) = v(g) \geq 0$. Normality makes b regular on the same open set. Thus the sheaf kernel is exactly $\mathcal{O}_X^p$. Frobenius is injective on the reduced structure sheaf and surjective onto this image sheaf, hence is an additive sheaf isomorphism.

The local primitives $\bar{G}_i|_X$ compute the connecting class as

$$\partial\nu = [(f_{ij}^p)] = \text{Fr}_*[(f_{ij})] = \text{Fr}_* \kappa_J.$$

It is nonzero by Proposition 6.1. This target is essential: the subsequent map to $H^1(X, \mathcal{O}_X)$ kills the connecting image by exactness. We have not asserted that the supersingular Frobenius operator on $H^1(X, \mathcal{O}_X)$ is invertible. Finally, X is a complete smooth geometrically integral genus-one curve by Theorem 4.1. A nonzero regular differential has an effective zero divisor of degree $2g - 2 = 0$. Its zero divisor is empty, proving the last assertion at every point of X . $\square$

Theorem 6.5 (Complete first-layer coefficient ideal). *At every point P of every original complete smooth finite fiber,*

$$\mathfrak{c}(\alpha_{mp})_P = (\pi, \tilde{H})_P, \quad (6.10)$$

where $\tilde{H}$ is any local lift of $H_p(T, J; \varepsilon_m)$. If h is a root of multiplicity e_h , the completed ideal, after an allowed unramified residue-field extension, is (π, z^{e_h}) with $\bar{z} = J - h$.

Proof. On an ordinary smooth fiber, (5.6) at $a = 1$ is $\bar{\alpha} = H dJ$, with nonzero H and nonvanishing dJ . The coefficient ideal is the unit ideal, which is $(\pi, \tilde{H})$ there.

On a supersingular fiber, choose a local lift j_i and extend dj_i to a basis (dj_i, θ) of the relative cotangent module at P . This is possible because dJ is a nonzero normal differential. Write (6.9) as

$$\alpha = (\hat{H}(j_i) + \pi A) dj_i + \pi B \theta.$$

Proposition 6.4 says that B is a unit at P . Consequently its two coefficients generate $(\hat{H}(j_i) + \pi A, \pi B) = (\pi, \hat{H}(j_i))$. All lifts of the same residue function differ by a multiple of π , so this is (6.10) with any $\tilde{H}$. The proof includes the terminal points because both (6.7) and the nonvanishing of ν hold on the complete fiber.

At a root of multiplicity e_h , write $H_p(T, J; \varepsilon_m) = (J - h)^{e_h} V(J)$ with $V(h) \neq 0$. Formal smoothness supplies a parameter z lifting $J - h$, together with a parameter along the curve, after the stated unramified extension and completion. The lift of V is a unit, and adjoining π to the ideal removes the choice of lift. The completed ideal is therefore exactly (π, z^{e_h}) . Ideal equalities obtained after these faithfully flat changes descend to the original local rings; their quantifiers are not enlarged to extra ramified states. $\square$

For an unramified lifted supersingular state, z maps into (π) , so the image of (π, z^{e_h}) is exactly (π) . The common order of the two coefficients of α_{mp} is one; on an ordinary state it is zero. Multiplication by p , whose uniformizer order is $p - 1$, gives respectively p and $p - 1$ for dI_{mp} . In particular, this is an equality of the complete original ideals, not merely a statement after adjoining π^2 .

7 Odd-prime first jets at all higher levels

The residue formula loses both state directions on a supersingular fiber. To retain their first vertical terms, we compare the original divided forms on a common square-zero thickening. The calculation

keeps the deformation inside each tame block as well as the rescaling between blocks. Neither a Hasse denominator nor a simple-root assumption occurs. This differs from working in a unit-root quotient after making the Hasse–Witt matrix invertible, as in [3]; the higher-Hasse framework discussed in [17] also has its own lower-level invertibility hypotheses. The first-jet identity needed here will follow directly from the original matrix.

7.1 The common thickening and the statement

In this section p is odd, $p \nmid m$, and the residue field k is finite. Use the coefficient rings and parameters of the main statements, and put

$$N = p^a, \quad \sigma = 1 + p + \cdots + p^{a-1}, \quad \ell = \sigma - 1, \quad T = \bar{t}_a^m, \quad J = I_{m,\eta}(x, y; \bar{t}_a).$$

Write $H = H_p(T, J; \varepsilon_m)$, where $\varepsilon_m = (-1)^{m+1}$. The cyclotomic valuation is $v_{\pi_i}(p) = p^{i-1}(p-1) \geq 2$. Thus $\mathcal{O}_i/(\pi_i^2)$ has characteristic p and a square-zero maximal ideal of length one. For $k = \mathbb{F}_{p^f}$ the roots of $X^{p^f} - X$ lift uniquely to this quotient, since its derivative is -1 . They form a coefficient field: sums and products remain roots, and reduction identifies their field with k . Every element therefore has a unique expression $c_0 + c_1\pi_i$, with $c_0, c_1 \in k$.

Identify the two quotients for $i = 1, a$ by

$$B = k[\epsilon]/(\epsilon^2), \quad \pi_1 \longmapsto \epsilon, \quad \pi_a \longmapsto \epsilon, \quad s_1, s_a \longmapsto \eta(1 + \epsilon), \quad (7.1)$$

where the displayed arrows mean the two specified maps to B . Choose $t_1 \in \mathcal{O}_1^\times$ whose image equals the entire image t of t_a in B , not just its residue. The quotient map is surjective, so such a choice exists. The four last center coordinates $1, t, t, s$ and the eight successive coordinate substitutions in (2.1) are then identical over B . They identify the two original open models with one smooth surface $\mathcal{U}_B$. This uses the actual blowup charts, not a general assertion that blowup commutes with every nonflat base change.

The comparison in (7.1) is not the natural cyclotomic embedding. For $a \geq 2$, that embedding sends π_1 to $\zeta_{p^a}^{p^{a-1}} - 1$, whose image in B is zero, not ϵ . The superscript [2] below always uses the specified common quotient and the matched time. Formula (5.1) shows on the torus that higher-order choices of the time lift do not affect the resulting form over B ; uniqueness on the terminal charts follows from Lemma 5.4.

Theorem 7.1 (Complete odd-prime first-jet factorization). *For every odd p , every m prime to p , and every $a \geq 2$ in the finite-residue-field setting above, the entire original open surface satisfies*

$$\alpha_{mp^a}^{[2]} = H^{\langle \sigma-1 \rangle} \alpha_{mp}^{[2]}. \quad (7.2)$$

Here $H^{\langle \sigma-1 \rangle}$ is the common power of any local regular lift of H . These powers glue even when J has no global lift. The equality includes all four terminal affine lines and does not require a smooth energy fiber, ordinarity, or a simple Hasse root.

We prove the theorem through block identities that also supply the characteristic-independent inputs to Section 8. In those algebraic identities the field containing η need not be finite, and a prime may be two whenever its square-zero quotient has that characteristic. The full model comparisons and geometric ideal statements retain their stated residue-field hypotheses.

7.2 Integral blocks and the full internal deformation

For the original matrix (1.1), set

$$\mathbf{B}_s(z) = A(s^{m-1}z; t) \cdots A(z; t), \quad r = s^m.$$

If s has exact order mN , then r has exact order N , and the original descending product is $\mathbf{B}_s(r^{N-1}z) \cdots \mathbf{B}_s(z)$.

Lemma 7.2 (Block insertion). *Before reduction, for every prime p and $N = p^a$,*

$$\alpha_{mN} = [z^{mN}] \operatorname{tr}(d\mathbf{B}_s(z) \mathbf{B}_s(r^{N-1}z) \cdots \mathbf{B}_s(rz)). \quad (7.3)$$

The right side is integral and includes all m insertions inside $d\mathbf{B}_s$; no additional factor m is present.

Proof. Apply the cyclic insertion argument of Lemma 5.1 to the N blocks. After rotating the differentiated block to the first position, substitute $w = r^j z$. The remaining block indices are $N-1, \dots, 1$, using $r^N = 1$, and the extracted coefficient is unchanged because $r^{jmN} = 1$. Thus the N block insertions give $dI_{mN} = N$ times the right side of (7.3). Division occurs in characteristic zero and leaves the displayed integral expression. $\square$

Now work on $R = B[x^{\pm 1}, y^{\pm 1}]$, keeping the actual $t \in B^\times$. For any polynomial with scalar or one-form coefficients define

$$\mathcal{P}_m \left(\sum_i b_i z^i \right) = \sum_{j \geq 0} b_{mj} Z^j. \quad (7.4)$$

This is a coefficient projection, not a ring homomorphism. Its valid multiplicative rule is

$$\mathcal{P}_m(Q(z^m)V(z)) = Q(Z)\mathcal{P}_mV(z). \quad (7.5)$$

Put $\mathbf{A}_j(z) = A(\eta^j z; t)$. Expanding every factor, including those inside the tame block, gives

$$\begin{aligned} \mathbf{B}_s &= \mathbf{B}_\eta(z; t) + \epsilon C(z), \\ C(z) &= \sum_{j=0}^{m-1} j \mathbf{A}_{m-1} \cdots \mathbf{A}_{j+1} (z \partial_z \mathbf{A}_j) \mathbf{A}_{j-1} \cdots \mathbf{A}_0. \end{aligned} \quad (7.6)$$

Empty products are identity matrices. Although C need only be specified modulo ϵ , the time in $\mathbf{B}_\eta(z; t)$ has not been replaced by $\bar{t}$. In particular, time derivatives of state differentials remain in this formula.

Lemma 7.3 (The deformed block trace). *Write $S_s = \operatorname{tr} \mathbf{B}_s$ and $D_s = \det \mathbf{B}_s$. Then*

$$\begin{aligned} S_s(z) &= S_*(z^m) + \epsilon V(z), & \mathcal{P}_m V &= 0, \\ S_*(Z) &= T_* + j_* Z + c_s Z^2, & T_* &= t^m, \quad j_* = [z^m] S_s, \quad c_s = s^{m(m-1)}, \\ D_s(z) &= d_s z^{3m}, & d_s &= s^{3m(m-1)/2}, \quad dS_* = Z dj_*, \quad dD_s = 0. \end{aligned} \quad (7.7)$$

The residue of j_ is J . For odd p the two state-constant leading coefficients are*

$$c_s = 1 + \epsilon m(m-1), \quad d_s = \epsilon_m \left(1 + \epsilon \frac{3m(m-1)}{2} \right). \quad (7.8)$$

The integer $3m(m-1)/2$ in these formulas is formed before reduction.

Proof. At $s = \eta$ the cyclic trace is invariant under $z \mapsto \eta z$. All nonmultiples of m vanish, since $\eta^i - 1$ is a unit for $m \nmid i$. At $z = 0$, the identities $\text{tr } A(0; t) = t$ and $\det A(0; t) = 0$ give $A(0; t)^m = t^{m-1} A(0; t)$. The constant trace coefficient is t^m . The highest block coefficient is $s^{m(m-1)} E$, because the highest coefficient of every original factor is a scalar multiple of $E = \text{diag}(1, 0)$ and $E^m = E$. Multiplying the original determinants gives $D_s = d_s z^{3m}$. These arguments remain valid over B with its actual unit time. Combining them with (7.6) proves (7.7). Finally, $\eta^{m(m-1)} = 1$ and $\eta^{3m(m-1)/2} = \varepsilon_m$: for odd m the exponent is a multiple of m , and for even m one uses $\eta^{m/2} = -1$. Expanding $s = \eta(1 + \epsilon)$ proves (7.8). $\square$

Lemma 7.4 (Square-zero nonresonant terms). *Under (7.7), put $D_*(Z) = d_s Z^3$. For any polynomial $\Phi(S, D)$ over the coefficient ring,*

$$\mathcal{P}_m(\Phi(S_s, D_s) dS_s) = \Phi(S_*, D_*) dS_*. \quad (7.9)$$

Proof. The square-zero expansion differs from the proposed right side only by the projections of

$$\epsilon \Phi_S(S_*(z^m), D_*(z^m)) V dS_*(z^m), \quad \epsilon \Phi(S_*(z^m), D_*(z^m)) dV.$$

Both are zero by (7.5), since $\mathcal{P}_m V = \mathcal{P}_m(dV) = 0$. Thus no nonresonant term contributes at this order. The resonant changes in j_*, c_s, d_s have not been discarded. $\square$

Return to odd p and define

$$F_N = [z^{mN}] \text{tr}(d\mathbf{B}_s \mathbf{B}_s^{N-1}), \quad f_* = S_*^2 - 4D_*, \quad H_* = [Z^{p-1}] f_*^{(p-1)/2}.$$

The universal polynomial identity for U_{N-1} proved in Lemma 5.2 holds over nonreduced rings as well. Equation (5.4), $dD_s = 0$, and Lemma 7.4 therefore give

$$F_N = [Z^{N-1}] f_*^{(N-1)/2} dj_*. \quad (7.10)$$

For completeness, the coefficient selection here includes all the deformed scalar coefficients. Put $Q_* = f_*^{(p-1)/2}$; its support lies in $[0, 2p-2]$, because $\deg f_* \leq 4$, and

$$f_*^{(N-1)/2} = \prod_{i=0}^{a-1} Q_*^{p^i}.$$

A contribution to Z^{p^a-1} has lowest index congruent to $p-1$ modulo p . In $[0, 2p-2]$ only $p-1$ has that congruence. Subtracting it and dividing the remaining degree by p repeats the argument at every index. Consequently

$$F_N = H_*^\sigma dj_*, \quad F_p = H_* dj_*. \quad (7.11)$$

Frobenius is applied only to scalar coefficients, never to a one-form. The functions T_*, j_*, H_* and the form dj_* still contain the full time jet. Since $\bar{H}_* = H$ and $p \mid \ell$ when $a \geq 2$, the power H_*^ℓ equals the power of any other lift of H .

7.3 The weighted commutator and the external terms

Lemma 7.5 (Weighted rank-two identity). *Let M, X be two-by-two matrices over a commutative ring of characteristic p , and let $N = p^a$. Then*

$$\sum_{j=1}^{N-1} j M^{N-1-j} X M^{j-1} = \text{ad}_M^{N-2}(X), \quad \text{ad}_M(X) = [M, X]. \quad (7.12)$$

For odd p , writing $S = \text{tr } M$ and $D = \det M$ gives

$$\sum_{j=1}^{N-1} j M^{N-1-j} X M^{j-1} = (S^2 - 4D)^{(N-3)/2} [M, X]. \quad (7.13)$$

No inverse of a determinant or discriminant is used.

Proof. In $\mathbb{F}_p[U, V]$ the geometric sum is

$$\sum_{j=0}^{N-1} U^{N-1-j} V^j = (U - V)^{N-1},$$

by $U^N - V^N = (U - V)^N$. This equality is first proved in the polynomial domain and then specialized, so cancellation is not being performed in the matrix coefficient ring. Differentiate with respect to V ; the right side has coefficient $-(N-1) = 1$. Substituting the commuting operators of left and right multiplication by M proves (7.12).

Over any commutative ring, Cayley–Hamilton gives $M^2 = SM - D \text{id}$ and $M^3 = (S^2 - D)M - SD \text{id}$. Substitution in the cubic commutator yields

$$\begin{aligned} \text{ad}_M^3(X) &= M^3 X - 3M^2 X M + 3M X M^2 - X M^3 \\ &= (S^2 - D)[M, X] - 3D[M, X] = (S^2 - 4D)[M, X]. \end{aligned}$$

For odd N , iterate this identity $(N-3)/2$ times to obtain (7.13). It remains valid for $N = 3$, with exponent zero, and for scalar matrices or vanishing discriminant. $\square$

Let $\mathbf{B}_0 = \mathbf{B}_\eta(z; \bar{t})$ be the residue block and set

$$\begin{aligned} \Gamma_m(z) &= \text{tr}(d\mathbf{B}_0[\mathbf{B}_0, z\partial_z \mathbf{B}_0]), \\ \gamma_m(Z) &= \mathcal{P}_m \Gamma_m(z), \quad f(Z) = (T + JZ + Z^2)^2 - 4\epsilon_m Z^3. \end{aligned} \quad (7.14)$$

The external speed is $r = s^m = 1 + m\epsilon$, whence

$$\mathbf{B}_s(r^j z) = \mathbf{B}_s(z) + \epsilon m j z \partial_z \mathbf{B}_0(z).$$

In the descending product of (7.3), an insertion at position j has $N-1-j$ unchanged blocks on its left and $j-1$ on its right. Thus (7.13) fixes the positive sign and gives

$$\begin{aligned} \alpha_{mN}^{[2]} &= F_N + \epsilon m \beta_N, \\ \beta_N &= [z^{mN}] \Gamma_m(z) f(z^m)^{(N-3)/2} = [Z^N] \gamma_m(Z) f(Z)^{(N-3)/2}. \end{aligned} \quad (7.15)$$

All other factors in an external correction are reduced because that correction already contains ϵ . This does not reduce the internal term F_N or its time jet.

Lemma 7.6 (Actual block support). *For the original residue block in any characteristic,*

$$\text{supp}_z \Gamma_m \subset [1, 6m-2], \quad \text{supp}_Z \gamma_m \subset \begin{cases} [1, 4], & m = 1, \\ [1, 5], & m \geq 2. \end{cases}$$

Proof. The highest coefficient of $\mathbf{B}_0$ is the state-constant matrix E , so $\deg d\mathbf{B}_0 \leq 2m-1$. The degree $4m$ coefficient of $[\mathbf{B}_0, z\partial_z \mathbf{B}_0]$ is $[E, 2mE] = 0$, while its constant term vanishes. Its degree is therefore at most $4m-1$. Multiplication by $d\mathbf{B}_0$ and projection to multiples of m give the asserted bounds. $\square$

The fifth possible resonant degree is essential for general m ; it cannot be removed by using the degree bound for a single original matrix. Put

$$Q = f^{(p-1)/2}, \quad G = \gamma_m f^{(p-3)/2}.$$

Lemma 7.6 gives the actual bounds

$$\text{supp } Q \subset [0, 2p - 2], \quad \text{supp } G \subset [1, 2p - 1]. \quad (7.16)$$

These also cover $p = 3$, when $G = \gamma_m$ and its upper bound is five. For $a \geq 2$, split the exponent in (7.15) to get

$$\beta_N = [Z^{p^a}]G(Z) \prod_{i=1}^{a-1} Q(Z)^{p^i}.$$

Every potentially contributing degree satisfies

$$n_0 + pn_1 + \cdots + p^{a-1}n_{a-1} = p^a, \quad 1 \leq n_0 \leq 2p - 1, \quad 0 \leq n_i \leq 2p - 2 \quad (i \geq 1).$$

Reduction modulo p forces $n_0 = p$, the only multiple of p in its permitted interval. The remaining degree, after subtraction and division by p , is $p^{a-1} - 1$. At each remaining index the only permissible value congruent to $p - 1$ is $p - 1$. The coefficient extraction is consequently exact:

$$\beta_N = H^{p+p^2+\cdots+p^{a-1}}\beta_p = H^\ell \beta_p. \quad (7.17)$$

Only Q , a scalar polynomial, is raised to Frobenius powers.

7.4 Global identity and the two-coefficient ideal

Completion of the proof of Theorem 7.1. The internal and external calculations give, on the common torus,

$$\alpha_{mN}^{[2]} = H_*^\sigma dj_* + \epsilon m H^\ell \beta_p, \quad \alpha_{mp}^{[2]} = H_* dj_* + \epsilon m \beta_p.$$

The two formulas use exactly the same s, t, B_s over B . They therefore retain both the tame-block deformation and the actual time term, including the change in dj_* .

If two local lifts of H differ by ϵb , their p th powers agree. Since $p \mid \ell$, their ℓ th powers agree too. These powers glue to a global regular $H^{(\ell)}$, and on the torus this function is H_*^ℓ . Multiplying the second displayed formula by it gives the first. The external factor m is identical at both levels and introduces no further factor into the comparison.

Both divided forms are globally regular by Proposition 5.3. Their difference after multiplication is therefore a regular one-form on $\mathcal{U}_B$, zero on the torus. The four terminal chart rings are the appropriate localizations of $B[u, v]$ by $1 + uv, t + uv, t + uv, s + uv$, and their torus intersections are obtained by inverting u . Lemma 5.4 makes restriction injective in the free cotangent module even over this nonreduced base. The difference vanishes on each of the four charts, proving the theorem on the whole open surface. No regularity assertion for β_p by itself is needed. $\square$

Corollary 7.7 (Exact odd-prime truncated ideal). *At every point of an original complete smooth finite energy fiber, for every $a \geq 2$ and odd p , one has*

$$\mathfrak{c}(\alpha_{mp^a}) + (\pi_a^2) = (\pi_a^2, \tilde{H}^\sigma, \pi_a \tilde{H}^{\sigma-1}). \quad (7.18)$$

The lift $\tilde{H}$ is arbitrary. If h is a root of multiplicity e_h of $H_p(T, h; \varepsilon_m)$, then, after an allowed unramified residue-field extension and completion, a coordinate z with $\bar{z} = J - h$ gives the ideal

$$(\pi_a^2, z^{e_h \sigma}, \pi_a z^{e_h(\sigma-1)}). \quad (7.19)$$

Proof. For a scalar g and a one-form ω in a free module, $\mathfrak{c}(g\omega) = g\mathfrak{c}(\omega)$, by taking both coefficients in a basis; g need not be a unit. Reduce the complete first-layer ideal of Theorem 6.5 to B and multiply its two generators by $H^{(\ell)}$, using Theorem 7.1. Taking the inverse image under reduction modulo π_a^2 gives (7.18). Thus the full-form comparison, rather than a tangential quotient alone, supplies both generators.

At a root of multiplicity e_h , the residue Hasse function is $(J - h)^{e_h} V(J)$ with $V(h) \neq 0$. Smoothness permits a local coordinate z lifting $J - h$, and the lift of $V(J)$ is a unit. Choose their product as the lift of H in (7.18), and remove unit factors from the generators. This proves (7.19) without assuming $e_h = 1$. The permissible local completions or finite unramified extensions do not alter the displayed original local identity. $\square$

On an ordinary smooth fiber the ideals in (7.18) are unit ideals. Along an allowed unramified state lifting a supersingular point, z has positive π_a -order, so (7.19) gives common order at least two for the divided form. It gives neither exact order two nor an untruncated higher-level critical ideal. The uniform state-order summary is given at the end of the next section.

8 Characteristic-two jets and mixed critical ideals

At the prime two, the first higher-level comparison starts at four blocks. Its base term contains a mixed contribution that the odd-prime formula cannot recover by substitution. We first calculate this term on the original torus, then identify its tangent component through the characteristic-four first layer. A local divisibility argument carries that identification to every point of the complete smooth fiber.

8.1 The four-block base and all higher block counts

Assume that m is odd, $a \geq 2$, and $N = 2^a$. The block calculations in this subsection allow the perfect residue field k of the main coefficient-ring setup; the geometric ideal conclusions below retain the finite-residue-field hypothesis. Set

$$B = \mathcal{O}_a/(\pi_a^2), \quad \epsilon = \pi_a \bmod \pi_a^2, \quad s = \eta(1 + \epsilon), \quad t = t_a \bmod \pi_a^2.$$

Use the original blocks

$$\mathbf{B}_s(z) = A(s^{m-1}z; t) \cdots A(z; t), \quad \mathbf{B}_0(z) = A(\eta^{m-1}z; \bar{t}) \cdots A(z; \bar{t}),$$

and put

$$j_* = [z^m] \operatorname{tr} \mathbf{B}_s(z), \quad J = I_{m, \eta}(x, y; \bar{t}), \quad T = \bar{t}^m, \quad S(Z) = T + JZ + Z^2.$$

The function j_* is the actual block coefficient on the torus, with residue J ; no global regular lift is asserted for it. Using the projection (7.4), define

$$\begin{aligned} \Gamma_m(z) &= \operatorname{tr}(d\mathbf{B}_0[\mathbf{B}_0, z\partial_z \mathbf{B}_0]), & \gamma_m &= \mathcal{P}_m \Gamma_m, \\ \beta_N &= [Z^N] \gamma_m(Z) S(Z)^{N-3}, \\ \chi_m &= J dJ + [z^{2m}] \operatorname{tr}(d\mathbf{B}_0 z \partial_z \mathbf{B}_0). \end{aligned} \tag{8.1}$$

In particular, χ_m is an ambient torus form defined by the original matrix, not by specifying its restriction to a fiber.

Theorem 8.1 (Characteristic-two first jets). *Identify $\mathcal{O}_2/(\pi_2^2)$ and $\mathcal{O}_a/(\pi_a^2)$ with $B = k[\epsilon]/(\epsilon^2)$ by sending their parameters to ϵ , and choose t_2 with the same entire image as t_a . On the common original torus,*

$$\alpha_{mN}^{[2]} = j_*^{N-1} dj_* + \epsilon \beta_N, \quad \beta_N = J^{N-4} \beta_4, \quad \beta_4 = T dJ + J^2 \chi_m. \quad (8.2)$$

Thus the four-block base is exactly

$$\alpha_{4m}^{[2]} = (j_*^3 + \epsilon T) dj_* + \epsilon J^2 \chi_m. \quad (8.3)$$

On the entire common original open surface, including its four complete terminal lines,

$$\alpha_{mN}^{[2]} = J^{(N-4)} \alpha_{4m}^{[2]}. \quad (8.4)$$

The multiplier is the common $(N-4)$ th power of local lifts of J , and is one when $N=4$. These form identities require no smoothness or ordinarity assumption on an energy fiber.

Proof. Step 1: the correct common quotient. The polynomial

$$\Phi_{2^a}(1+X) = (1+X)^{2^{a-1}} + 1$$

is Eisenstein at two: its constant term is two, all other nonleading coefficients are divisible by two, and the constant term is not divisible by four. Hence π_a is a uniformizer, and $v_{\pi_a}(2) = 2^{a-1} \geq 2$. The quotient B therefore has characteristic two. The image of $\mathcal{O}_0$ factors through $\mathcal{O}_0/(2) = k$ and gives its coefficient field, and each element of B has a unique form $u + v\epsilon$. This proves the stated dual-number description, also at $a=2$.

The time lift t_2 exists by surjectivity of the quotient map. The identical parameters $1, t, t, s$ in the original eight blowup charts give the common model, just as in (7.1). This is not the natural embedding $i \mapsto \zeta_{2^a}^{2^{a-2}}$: when $a > 2$ it sends $i-1$ to zero in B . Nor can this quotient be identified with that of height one, where $\pi_1 = -2$ and $\mathcal{O}_1/(\pi_1^2)$ has characteristic four.

Step 2: integral insertion and the internal deformation. Lemma 7.2 applies before reduction for $N = 2^a$ and yields the same integral block expression (7.3). Its normalization divides by N , not by mN ; all m differentiated factors are inside $d\mathbf{B}_s$. The full expansion (7.6) remains valid over the present B . In particular, every internal spectral perturbation and the actual time jet are retained.

Specializing Lemma 7.3 gives

$$\begin{aligned} S_s &= \text{tr } \mathbf{B}_s = S_*(z^m) + \epsilon V(z), & \mathcal{P}_m V &= 0, \\ S_*(Z) &= t^m + j_* Z + Z^2, & D_s &= s^{3m(m-1)/2} z^{3m}, & dD_s &= 0. \end{aligned} \quad (8.5)$$

The highest trace coefficient is exactly one in B , because $s^{m(m-1)} = 1$ and $m(m-1)$ is even. The determinant coefficient may still have a nonzero first-order term; it has not been deleted, but its state differential is zero. When $m=1$, the sum of internal spectral perturbations is empty, while the time dependence in $A(z; t)$ remains.

Step 3: the characteristic-two internal term. The rank-two recurrence (5.4) and the universal identity $U_{N-1}(S, D) = S^{N-1}$ in characteristic two give

$$F_N := [z^{mN}] \text{tr}(d\mathbf{B}_s \mathbf{B}_s^{N-1}) = [z^{mN}] S_s^{N-1} dS_s.$$

This identity is polynomial, so it remains true with square-zero coefficients. Lemma 7.4, applied with $\Phi(S, D) = S^{N-1}$, eliminates precisely the two nonresonant error terms

$$\epsilon(N-1) S_*^{N-2} V dS_*, \quad \epsilon S_*^{N-1} dV,$$

and no resonant deformation. Thus

$$F_N = [Z^{N-1}]S_*(Z)^{N-1}dj_*.$$

Now $S_*^{N-1} = \prod_{i=0}^{a-1} S_*^{2^i}$ and the allowed index in each original quadratic factor is 0, 1, 2. To contribute to $2^a - 1$, the lowest index must be odd, hence equal to one. After subtraction and division by two, every remaining index is again forced to one. Consequently

$$F_N = j_*^{N-1}dj_*. \quad (8.6)$$

The coefficient dj_* includes the variation of the state differential caused by any nonzero first-order part of t .

Step 4: the characteristic-two weighted commutator. For a two-by-two matrix M in characteristic two, write $S_M = \text{tr } M$ and $D_M = \det M$. Cayley–Hamilton gives $M^2 = S_MM + D_M \text{id}$, and hence

$$[M, [M, X]] = M^2X + XM^2 = S_M[M, X].$$

Combining this relation with the characteristic-independent operator identity (7.12), for $N \geq 4$, gives the polynomial equality

$$\sum_{j=1}^{N-1} jM^{N-1-j}XM^{j-1} = \text{ad}_M^{N-2}(X) = S_M^{N-3}[M, X]. \quad (8.7)$$

It does not divide by S_M and remains valid when $S_M = 0$.

Because m is odd, $r = s^m = 1 + \epsilon$, so each external block has expansion

$$B_s(r^j z) = B_s(z) + \epsilon j z \partial_z B_0(z).$$

Expand the descending product in (7.3). In a term already multiplied by ϵ , the other factors and the differential may be reduced. Equation (8.7) therefore yields

$$\alpha_{mN}^{[2]} = F_N + \epsilon[Z^N]\gamma_m S^{N-3} = F_N + \epsilon\beta_N. \quad (8.8)$$

The external speed is the residue of m , namely one. This is distinct from the integral normalization in Step 2.

Step 5: actual support and binary selection. Lemma 7.6 applies to Γ_m , so γ_m is supported in degrees one through five, with upper bound four when $m = 1$. The product $\gamma_m S$ is consequently supported in $[1, 7]$. In characteristic two,

$$\gamma_m S^{N-3} = (\gamma_m S) \prod_{i=2}^{a-1} S^{2^i}.$$

To contribute to degree $N = 2^a$, the first index must be divisible by four. The only such index in $[1, 7]$ is four. Subtracting four and dividing by four leaves the target $2^{a-2} - 1$. Each remaining index lies in $\{0, 1, 2\}$, and the same binary argument forces every one to equal one. Its scalar coefficient is $J^{4+8+\dots+2^{a-1}} = J^{N-4}$. Thus

$$\beta_N = J^{N-4}[Z^4](\gamma_m S) = J^{N-4}\beta_4. \quad (8.9)$$

For $a = 2$ the subsequent product is empty, so this includes the base case. No Frobenius power is taken of a one-form.

Step 6: the actual four-block coefficient. Put $M = \mathbf{B}_0(z)$, $R_z = z\partial_z M$, $\mathcal{S} = \text{tr } M = S(z^m)$, and $\mathcal{D} = \det M = z^{3m}$. The determinant sign is one because m is odd. Differentiating $M^2 + \mathcal{S}M + \mathcal{D} \text{id} = 0$ in a state direction, using $d\mathcal{D} = 0$, gives

$$M dM + dM M = (d\mathcal{S})M + \mathcal{S} dM.$$

Multiplication by R_z and cyclic trace, with just one differential factor in every term, give

$$\Gamma_m = d\mathcal{S} \text{tr}(MR_z) + \mathcal{S} \text{tr}(dM R_z).$$

The adjugate identity $\text{adj}(M) = \mathcal{S} \text{id} + M$ also implies $z\partial_z \mathcal{D} = \mathcal{S}(z\partial_z \mathcal{S}) + \text{tr}(MR_z)$. Therefore

$$\begin{aligned} \Gamma_m &= (z\partial_z \mathcal{D})d\mathcal{S} + \mathcal{S}K_m, \\ K_m &= d\mathcal{S}(z\partial_z \mathcal{S}) + \text{tr}(dM R_z). \end{aligned} \tag{8.10}$$

This calculation uses the actual state-constant determinant; it is not an identity without extra terms for arbitrary state-dependent determinants.

Since m and $3m$ are odd,

$$d\mathcal{S} = z^m dJ, \quad z\partial_z \mathcal{S} = Jz^m, \quad z\partial_z \mathcal{D} = z^{3m}.$$

It follows that

$$\Gamma_m = z^{4m} dJ + S(z^m)K_m, \quad K_m = J dJ z^{2m} + \text{tr}(d\mathbf{B}_0 z\partial_z \mathbf{B}_0). \tag{8.11}$$

Here $\deg d\mathbf{B}_0 \leq 2m - 1$. In characteristic two the derivative of its leading term $z^{2m}E$ also vanishes, so $\deg(z\partial_z \mathbf{B}_0) \leq 2m - 1$. The trace term in (8.11) has degree at most $4m - 2$ and zero constant coefficient. Since $2m \leq 4m - 2$ for all $m \geq 1$, the same assertions hold for K_m . Thus $\kappa_m = \mathcal{P}_m K_m$ is supported in degrees one through three, with upper bound two when $m = 1$, and $[Z^2]\kappa_m = \chi_m$. Projection of (8.11) yields

$$\gamma_m = Z^4 dJ + S\kappa_m.$$

Because $S^2 = T^2 + J^2 Z^2 + Z^4$ and $[Z^0]\kappa_m = [Z^4]\kappa_m = 0$, we obtain

$$\beta_4 = [Z^4](\gamma_m S) = T dJ + J^2 [Z^2]\kappa_m = T dJ + J^2 \chi_m.$$

Together with (8.6), (8.8), and $\epsilon dJ = \epsilon dj_*$, this proves (8.2) and (8.3).

Step 7: gluing the multiplier across the terminal lines. If $N > 4$, then $N - 4$ is even. For any two local regular lifts $j, j + \epsilon f$ of J ,

$$(j + \epsilon f)^2 = j^2, \quad (j + \epsilon f)^{N-4} = j^{N-4}.$$

The powers therefore glue to the regular function $J^{(N-4)}$. For $N = 4$ use the function one. On the common torus, (8.9) and (8.6) give

$$\alpha_{mN}^{[2]} = j_*^{N-4}(j_*^3 dj_* + \epsilon \beta_4).$$

The form in parentheses is the four-block form for the same s, t over B . Both sides of (8.4) are regular by Proposition 5.3, and restriction is injective on each of the four nonreduced terminal charts by Lemma 5.4. The torus identity thus extends across every terminal line. This proof does not assume that χ_m , β_4 , or j_* separately extends to the whole surface. $\square$

8.2 Identification with the actual first-layer tangent form

For the geometric conclusions, take the finite residue field k and the permitted finite unramified extensions in the main setup. Assume first that the complete scheme fiber $X = (J = 0)$ is smooth. By Theorem 4.1 and Proposition 4.2, it is a geometrically integral genus-one curve and is precisely a smooth supersingular fiber, since $H_2(T, h) = h$. Let ν be the actual nowhere-vanishing first-layer form of Proposition 6.4.

Lemma 8.2 (Characteristic-four comparison). *On the torus intersection with X , the form defined in (8.1) satisfies*

$$\chi_m|_{\Omega_X^1} = \nu. \quad (8.12)$$

In particular, its tangent restriction has the specified regular nonzero extension to the complete X . This does not assert ambient regularity of χ_m on the entire residue surface.

Proof. Use a separate first-layer coefficient ring with $\pi_1 = -2$, $s_1 = -\tilde{\eta}$, and a unit time t_1 whose residue is $\bar{t}_a$. Only this residue is being matched; no characteristic-four quotient is identified with the higher dual-number quotient. Write

$$B_s(z) = A(s_1^{m-1}z; t_1) \cdots A(z; t_1), \quad S_s = \text{tr } B_s, \quad j_* = [z^m]S_s,$$

using j_* here for this first-layer block coefficient, and put $\delta_1 = s_1^{3m(m-1)/2}$. As $s_1^m = -1$, integral block insertion gives

$$\alpha_{2m} = [z^{2m}] \text{tr}(dB_s(z)B_s(-z)).$$

In the characteristic-four ring $\mathcal{O}_1/(\pi_1^2)$, the monomial expansion of $-1 = 1 + \pi_1$ gives

$$B_s(-z) = B_s(z) + \pi_1 z \partial_z B_s(z).$$

The integer identity $\text{tr}(dM M) = (\text{tr } M)d(\text{tr } M) - d(\det M)$ now computes the internal contribution; it does not require division by two. The determinant of B_s is state-constant. Its highest trace coefficient is exactly one, since $s_1^{m(m-1)} = 1$. Its nonresonant trace coefficients are divisible by π_1 , by cyclic invariance after reduction. In the coefficient $[z^{2m}]S_s dS_s$, a term with exactly one nonresonant factor is still nonresonant, and a term with two such factors has π_1^2 . Thus this coefficient is $j_* dj_*$ modulo π_1^2 .

Put

$$K_0 = [z^{2m}] \text{tr}(dB_0 z \partial_z B_0).$$

Reducing only the external correction, already multiplied by π_1 , gives

$$\alpha_{2m} \equiv j_* dj_* + \pi_1 K_0 \pmod{\pi_1^2}. \quad (8.13)$$

On the other hand, the original prime trace polynomial (6.3), with Lemma 6.2, has, at $p = 2$ and odd m ,

$$F(h) = -[Z^2]((t_1^m + hZ + Z^2)^2 - 2\delta_1 Z^3) = -h^2 - 2t_1^m.$$

Consequently the actual torus function $G = (I_{2m} - F(j_*))/(2\pi_1)$ is regular and satisfies

$$\alpha_{2m} = -j_* dj_* + \pi_1 dG. \quad (8.14)$$

Compare (8.13) and (8.14) before reducing two to zero. The intermediate term is $2j_* dj_* = -\pi_1 j_* dj_*$. The identification $\pi_1 \Omega / \pi_1^2 \Omega \simeq \Omega \otimes k$ therefore gives

$$d\bar{G} = K_0 - JdJ = K_0 + JdJ = \chi_m.$$

Only the last equality uses characteristic two. By the actual first-layer construction in Proposition 6.4, $d(\bar{G}|_X) = \nu$; changing the local lift adds a function whose restriction differs by a square and hence does not change this differential. This proves (8.12) with the original tangent class, not an unspecified scalar multiple of it. $\square$

8.3 Local difference quotients and the two divisibility steps

Return to height two and its characteristic-two quotient $B = \mathcal{O}_2/(\pi_2^2)$. On local opens covering X , choose regular lifts j_i of J . The residue formula (5.6) is $\bar{\alpha}_{4m} = J^3 dJ$. Since the relative cotangent module is locally free over the smooth B -surface, multiplication by ϵ identifies its residue module with its ϵ -multiple. It therefore defines a unique regular residue one-form

$$\beta_i = \frac{\alpha_{4m}^{[2]} - j_i^3 dj_i}{\epsilon} \bmod \epsilon. \quad (8.15)$$

This denotes a unique residue class, not a unique numerical division inside the dual-number ring.

If $j_j = j_i + \epsilon f$, the full two-direction expansion is

$$(j_i + \epsilon f)^3 d(j_i + \epsilon f) = j_i^3 dj_i + \epsilon(j_i^2 f dj_i + j_i^3 df),$$

so

$$\beta_j - \beta_i = -J^2 \bar{f} dJ - J^3 d\bar{f}. \quad (8.16)$$

On the intersection of such an open with the torus, use j_* from (8.3). Equations (8.3) and (8.16) show that $\beta_i - TdJ$ is divisible by J^2 , and that its quotient differs from χ_m by a form of type $\bar{f} dJ + Jd\bar{f}$.

Lemma 8.3 (Divisibility at every complete-fiber point). *For every $P \in X$, including every point on the four terminal lines, the form*

$$\rho_i = (\beta_i - TdJ)/J^2 \quad (8.17)$$

is regular near P , and its tangent restriction to X equals ν . Its tangent coefficient is therefore a unit at P .

Proof. The smooth geometrically integral scheme fiber $X = (J = 0)$ is a reduced prime Cartier divisor in the smooth residue surface. Thus $R_P = \mathcal{O}_{\mathcal{U}_k, P}$ is a domain, (J) is prime, and dJ is a nonzero conormal parameter at P . The generic point of X lies on the original torus. Indeed, otherwise the integral complete genus-one curve would lie in one of the four terminal affine lines, which cannot contain such a complete curve.

The torus calculation consequently proves J^2 divisibility at the generic point of this prime divisor. In a local free frame let $b \in R_P$ be any coefficient of $\beta_i - TdJ$. There are $u \notin (J)$ and $g \in R_P$ with $ub = J^2 g$. Primality first gives $b = Jb_1$. Cancelling J in the domain gives $ub_1 = Jg$, and primality a second time gives $b_1 \in (J)$. Hence $b \in J^2 R_P$. Applying these two divisions to each coefficient proves the asserted regularity.

On the torus intersection, (8.16) shows that the tangent restriction of ρ_i is that of χ_m , since both dJ and J disappear in Ω_X^1 . Lemma 8.2 identifies it with ν . Both are now regular along the complete integral X , so their equality extends to every point. The first-layer nowhere-vanishing property of ν proves the unit assertion. No global regularity of the ambient χ_m has been assumed. $\square$

Theorem 8.4 (The characteristic-two mixed ideals). *For every odd m , every $a \geq 2$, and every point P of an original complete smooth finite energy fiber in the stated finite-residue-field setting,*

$$\mathfrak{c}(\alpha_{mN})_P + (\pi_a^2) = (\pi_a^2, j^{N-1} + \pi_a \tilde{T} j^{N-4}, \pi_a j^{N-2})_P. \quad (8.18)$$

Here j is any local lift of J , and $\tilde{T}$ is any constant unit lift of T . In particular, at height two the ideal is

$$(\pi_2^2, j^3 + \pi_2 \tilde{T}, \pi_2 j^2). \quad (8.19)$$

These are exact truncated ideals; all two state coefficients are retained.

Proof. At a point of $X = (J = 0)$, extend dj_i to a local relative cotangent basis (dj_i, θ) . This is possible because dJ is a nonzero conormal coordinate. From (8.15) and Lemma 8.3, write a lift of ρ_i as $A_i dj_i + B_i \theta$; its tangent coefficient B_i is a unit. The actual local form is

$$\alpha_{4m}^{[2]} = (j_i^3 + \epsilon \tilde{T}) dj_i + \epsilon j_i^2 (A_i dj_i + B_i \theta).$$

Its coefficient ideal is therefore

$$(j_i^3 + \epsilon \tilde{T} + \epsilon j_i^2 A_i, \epsilon j_i^2 B_i) = (j_i^3 + \epsilon \tilde{T}, \epsilon j_i^2). \quad (8.20)$$

Taking the inverse image from the square-zero quotient proves (8.19) in the original local ring. Changing j_i by ϵf changes its cube by $\epsilon J^2 \bar{f}$, already in the second generator; changing $\tilde{T}$ changes only a square-zero error. This verifies the arbitrary-lift assertion at height two.

For height a , use the common quotient with exactly matched time in Theorem 8.1. Multiplication by $J^{(N-4)}$ multiplies both coefficients in (8.20), regardless of whether that scalar is a unit. In a local lift it is j_i^{N-4} , giving the two generators in (8.18) after taking the inverse image modulo π_a^2 . For another lift, j^{N-1} changes by $\epsilon J^{N-2} f$, contained in the other generator, and ϵj^{N-4} is unchanged. Thus the formula has the stated lift independence at every height.

On an ordinary smooth fiber $J = h \neq 0$, (5.6) reads $\bar{\alpha}_{mN} = J^{N-1} dJ$ with a unit coefficient. The actual coefficient ideal is then the unit ideal, and the right side of (8.18) is also a unit ideal. This treats all smooth finite fibers, not just $X = (J = 0)$. $\square$

8.4 The base parameter, transverse length, and state orders

Corollary 8.5 (The height-two truncated quotient). *At a closed point P of a smooth supersingular fiber, after the allowed finite unramified extension and completion, there are coordinates w, z , with $\bar{z} = J$ and w along X , such that*

$$\begin{aligned} \frac{\hat{\mathcal{O}}_{\mathcal{U}_2, P}}{\mathfrak{c}(\alpha_{4m}) + (\pi_2^2)} &\simeq \frac{k(P)[[w, z, \epsilon]]}{(\epsilon^2, z^3 + \epsilon T, \epsilon z^2)} \\ &\simeq k(P)[[w, z]]/(z^5), \quad \pi_2 \mapsto z^3/T. \end{aligned} \quad (8.21)$$

The transverse length of this truncated quotient is five. It is not the length of the full critical quotient.

Proof. Relative smoothness and the nonzero conormal dJ allow z to be a lift of J in a completed coordinate system. Modulo π_2^2 , the complete local ring is $k(P)[[w, z, \epsilon]]/(\epsilon^2)$. The ideal (8.19) gives the first quotient in (8.21). Since $T \neq 0$, the mixed relation eliminates $\epsilon = z^3/T$. The relation $\epsilon z^2 = 0$ becomes $z^5 = 0$, while $\epsilon^2 = 0$ becomes $z^6 = 0$, already a consequence. Thus the quotient is free of rank five over $k(P)[[w]]$, with basis $1, z, z^2, z^3, z^4$, and has the displayed action of the original base parameter. $\square$

The remaining coordinate w makes this a transverse length, not a total Artin length at the closed point. Nor does the calculation assert flatness or determine the untruncated critical quotient. After a further permitted finite unramified extension, choose a residue unit c_0 with $c_0^3 = T$. The coordinate change $z = c_0 z'$ normalizes the coefficient in the base action to $\pi_2 = z'^3$. Thus T is not a new formal modulus. The calculation determines its original mixed term and base action, which cannot be obtained merely from the number five. Concrete characteristic-two elliptic lifts and height-two formal groups already occur in [13, §3.5]; their existence does not identify the two original state coefficients calculated here.

Corollary 8.6 (Orders along the allowed states). *Consider a state over the stated cyclotomic DVR, after at most an allowed finite unramified extension, whose residue is a point of a complete smooth finite fiber. Let ord_{π_a} of a one-form along that state mean the minimum order of its two coefficients in a local relative cotangent frame. The exact values and lower bounds are those stated in Table 1. On a supersingular fiber the divided order is exactly one at height one for every prime, and also at height two for $p = 2$. For odd p at height at least two, and for $p = 2$ at height at least three, it is at least two. The corresponding original orders are exactly p , exactly five, or at least $a\varphi(p^a) + 2$, respectively. On an ordinary smooth fiber the orders are exactly zero for the divided form and $a\varphi(p^a)$ for the original differential.*

Proof. At height one the full ideal of Theorem 6.5, including its multiple-root form, has image (π_1) along a supersingular lifted state: the lifted transverse parameter lies in (π_1) . Thus the divided order is exactly one. For odd p and $a \geq 2$, the ideal (7.19) has image contained in (π_a^2) , which proves only the stated lower bound.

At $p = 2, a = 2$, evaluation of a lift j of J lies in (π_2) . The term $j^3 + \pi_2 \tilde{T}$ has order exactly one, whereas $\pi_2 j^2$ has order at least three. The image of the original ideal after adding (π_2^2) is therefore (π_2) . The original ideal itself must have order one, since an ideal of order at least two cannot acquire an order-one element by adding (π_2^2) . For $a \geq 3, N \geq 8$ and both nonconstant generators in (8.18) evaluate to order at least two. The truncated information consequently gives order at least two, not equality.

On an ordinary fiber the residue multiplier is nonzero and dJ is a coordinate differential, so one coefficient is a unit. Finally, $v_{\pi_a}(p) = \varphi(p^a)$ and $dI_{mp^a} = p^a \alpha_{mp^a}$. Adding $a\varphi(p^a)$ to each divided order gives the original orders, in particular p at height one and five at $(p, a) = (2, 2)$. $\square$

These pointwise lower bounds do not say that a higher-level form is globally divisible by π_a^2 on the surface: the residue form is nonzero on the ordinary locus. The full first-layer ideal and the higher-level ideals modulo π_a^2 answer different thickness questions. The present results do not determine full higher critical ideals, exact higher supersingular orders beyond the cases stated, singular-energy ideals, or orders after additional ramified state extensions. All identities concern the original eight-blowup model and its complete terminal charts.

References

- [1] Jeffrey D. Achter and Everett W. Howe. Hasse–Witt and Cartier–Manin matrices: A warning and a request. In Y. Aubry, E. W. Howe, and C. Ritzenthaler, editors, *Arithmetic Geometry: Computation and Applications*, volume 722 of *Contemporary Mathematics*, pages 1–18. American Mathematical Society, 2019. doi: 10.1090/conm/722/14534. URL <https://arxiv.org/abs/1710.10726v5>. Corrected author version arXiv:1710.10726v5, 7 February 2020.
- [2] Arnaud Beauville. Jacobiennes des courbes spectrales et systèmes hamiltoniens complètement intégrables. *Acta Mathematica*, 164:211–235, 1990. doi: 10.1007/BF02392754.
- [3] Frits Beukers and Masha Vlasenko. Dwork crystals I. *International Mathematics Research Notices*, 2021(12):8807–8844, 2021. doi: 10.1093/imrn/rnaa119. URL <https://arxiv.org/abs/1903.11155v4>.
- [4] Serge Cantat and Igor Dolgachev. Rational surfaces with a large group of automorphisms. *Journal of the American Mathematical Society*, 25(3):863–905, 2012. doi: 10.1090/S0894-0347-2012-00732-2.
- [5] Pierre Deligne and Luc Illusie. Relèvements modulo p^2 et décomposition du complexe de de Rham. *Inventiones Mathematicae*, 89:247–270, 1987. doi: 10.1007/BF01389078.
- [6] Taylor Dupuy and David Zureick-Brown. Deligne–Illusie classes as arithmetic Kodaira–Spencer classes. *Journal de Théorie des Nombres de Bordeaux*, 31(2):371–383, 2019. doi: 10.5802/jtnb.1086.
- [7] Taylor Dupuy, Eric Katz, Joseph Rabinoff, and David Zureick-Brown. Total p -differentials on schemes over $\mathbf{Z}/p^2$. *Journal of Algebra*, 524:110–123, 2019. doi: 10.1016/j.jalgebra.2019.01.003. URL <https://arxiv.org/abs/1712.09487v1>.
- [8] Mark Gross, Paul Hacking, and Sean Keel. Moduli of surfaces with an anti-canonical cycle. *Compositio Mathematica*, 151(2):265–291, 2015. doi: 10.1112/S0010437X14007611.
- [9] Rei Inoue, Pol Vanhaecke, and Takao Yamazaki. Algebraic integrable systems related to spectral curves with automorphisms. *Journal of Geometry and Physics*, 87:198–216, 2015. doi: 10.1016/j.geomphys.2014.07.008. URL <https://arxiv.org/abs/1312.4208v1>.
- [10] Nalini Joshi and S. B. Lobb. Singular dynamics of a q -difference Painlevé equation in its initial-value space. *Journal of Physics A: Mathematical and Theoretical*, 49:014002, 2016. doi: 10.1088/1751-8113/49/1/014002. URL <https://arxiv.org/abs/1407.1961v3>.
- [11] Nalini Joshi and Pieter Roffelsen. Arithmetic dynamics of a discrete Painlevé equation. *Journal of Physics A: Mathematical and Theoretical*, 59:195201, 2026. doi: 10.1088/1751-8121/ae67bf. URL <https://arxiv.org/abs/2508.18578v2>. Author version: arXiv:2508.18578v2.
- [12] Peter Koroteev and Andrey Smirnov. On the quantum K-theory of quiver varieties at roots of unity. *International Mathematics Research Notices*, 2026(14):rnag153, 2026. doi: 10.1093/imrn/rnag153. URL <https://arxiv.org/abs/2412.19383v4>.
- [13] J. Lubin and J. Tate. Formal moduli for one-parameter formal Lie groups. *Bulletin de la Société Mathématique de France*, 94:49–59, 1966. doi: 10.24033/bsmf.1633.

- [14] Kanau Shimada. Arithmetic Kodaira–Spencer class and Frobenius liftings via Frobenius–Witt cotangent complex. arXiv:2608.21772v1, 2026. URL <https://arxiv.org/abs/2608.21772v1>.
- [15] The Stacks Project Authors. The Stacks Project, 2026. URL <https://stacks.math.columbia.edu/>. Online reference, accessed 9 September 2026; individual tags cited in the text.
- [16] Masha Vlasenko. Higher Hasse–Witt matrices. *Indagationes Mathematicae*, 29:1411–1424, 2018. doi: 10.1016/j.indag.2018.07.004. URL <https://arxiv.org/abs/1605.06440v3>. Theorem comparisons use author version arXiv:1605.06440v3.
- [17] Masha Vlasenko. Cohomology and congruences. arXiv:2412.13313v1, 2024. URL <https://arxiv.org/abs/2412.13313v1>. Author notes; the manuscript title reads “Congruences and cohomology”.